The Load-Bearing Test: is AI safety a tax on capability, or part of what holds it up?

Empirical work · Experimental paper · Paper VIII

Recursive Dynamics · The ARC Theory · bearing on Law V, the ARC Embedding Law, a law under test · The Eden Protocol

Michael Darius Eastwood

Independent Researcher, London, United Kingdom
ORCID 0009-0004-3222-7442 michael@michaeldariuseastwood.com

Abstract

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. Safety is often taken to cost capability, which in a race to be first gives developers a reason to defer it. Is the cost real, or can safety be part of what holds capability up? Three separate experiments look for it. In the first, a Darwin Gödel Machine on DeepSeek V3.2 evolved 75 agents under three selection conditions, scored by GPT-5.4, a blinded judge from another provider: no difference was detected between the conditions (p from 0.28 to 0.74, Mann-Whitney U), so the safety gate cost no measurable capability and brought no measurable benefit. In the second, LoRA fine-tuning of Qwen 2.5 3B Instruct under capability, safety-proxy and entangled losses, with 9 and then 295 training examples, left every fine-tuned condition below the base model on capability (7.68 against at best 4.00 at the larger scale), so the entangled objective's capability cost could not be determined; on the paper's candidate account, the base model's existing RLHF training is too strong for fine-tuning at this scale. In the third, a gated self-modification simulation in a toy system, the unconstrained arm gained capability and lost safety (here, retention of earlier tasks: 0.718 and 0.933 against the static baseline's 0.687 and 0.956), while the gated arm stayed above the baseline on capability (0.696 ± 0.014) with safety 0.947 ± 0.025, three seeds an arm and no difference tested; all of it is simulation output, prediction-side and never an empirical outcome. One null, one inconclusive and one simulation result leave the safety-capability trade-off unproven at the scale tested, and whether embedded safety brings a measurable benefit open. The paper informs Law V, the ARC Embedding Law, a law under test, and decides none of its propositions.

Version 3.8 sets the paper on the programme's presentation standard; calls its experiments separate and its judge blinded, from another provider; states what each experiment's safety measure is, and describes Experiment 1's gate and Experiment 3's drag control as the code runs them; reports Experiment 1's null as no difference detected, and notes that no arm varied recursive depth; corrects the epigraphs, Figure 1 and the dating of the code; and brings the laws' status up to date. Appendix B lists every version and correction, with its date.

Paper VIII of the ARC Theory, within Recursive Dynamics: the Eden Protocol's load-bearing test, at three levels, with one null, one inconclusive and one simulation result. It informs Law V, the ARC Embedding Law, without deciding it: the theory registration records its weight-level experiment under P21, one of Law V's two propositions, as inconclusive and not adverse, and, since no arm varied recursive depth, as not testing what P21 claims (Section 11 gives the laws' status). Working paper, version 3.8, 9 November 2026; first published 18 March 2026; Appendix B lists every version and correction. Recorded on OSF as *Paper VIII · The ARC Theory · Law V, the ARC Embedding Law · The Eden Protocol · The Load-Bearing Test: Whether Safety and Capability Are Structurally Entangled*, DOI 10.17605/OSF.IO/7YJ4E. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e. Licence: CC BY 4.0.

Contents

1 Introduction **4**
1.1 The safety-capability trade-off assumption . . . 5
1.2 The alternative hypothesis . . . 5
1.3 This paper's contribution . . . 5

2 Common framework **6**
2.1 The three conditions . . . 6
2.2 Why these names . . . 6
2.3 Shared metrics . . . 6

3 Experiment 1: self-improving AI with a judge from another provider **7**
3.1 Design . . . 7
3.2 Results . . . 8
3.3 Why this is a null result . . . 9

4 Experiment 2: weight-level embedding **10**
4.1 Design . . . 10
4.2 Training loss convergence . . . 11
4.3 Base model baseline . . . 11
4.4 Fine-tuned evaluation results . . . 12
4.5 The removal test (v1 only) . . . 13
4.6 What the removal test does and does not prove (v1) . . . 14
4.7 The removal gradient (v1) . . . 15

5 Experiment 3: gated self-modification simulation **16**
5.1 Design . . . 16
5.2 Results . . . 17
5.3 The Babylon fingerprint . . . 18
5.4 The Drag Control comparison . . . 18

6 Convergent analysis **20**
6.1 Three-level convergence . . . 20
6.2 Connection to Papers V, VI, and VII . . . 20
6.3 The ARC Principle connection . . . 21
6.4 The nearest external work . . . 24
6.5 Same-class versus cross-class correctors: a falsifiable prediction . . . 24
6.6 A second hypothesis from the same mechanism . . . 24
6.7 Terminology note: α across the programme's eras . . . 25

7 Limitations **25**
7.1 Scale . . . 25
7.2 The weight-level experiment . . . 25
7.3 AI-as-judge . . . 26
7.4 Statistical power . . . 26
7.5 Blinding and laundering without consensus scoring . . . 27
7.6 What we do not claim . . . 27
7.7 Falsifiability: what would overturn these results . . . 28

8 Future work **28**
8.1 Weight experiment redesign . . . 29
8.2 Removal gradient at adequate scale . . . 29
8.3 DGM with more mutable foundation models . . . 29
8.4 Cross-architecture replication . . . 29
8.5 Red-teaming . . . 29
8.6 Verification tax quantification . . . 30

9 Conclusion **30**

10 Related work **31**

11 What this paper does and does not show **32**

References **34**

A Measures as the published code computes them **38**

B Version history and corrections **38**
B.1 Version 3.8 . . . 38
B.2 Earlier versions, as printed . . . 44
B.3 Dated notes, as printed . . . 47
B.4 Version 3.7 wording, as printed . . . 48
B.5 Layout . . . 60

C Priority record **60**
C.1 Dated records . . . 61
C.2 Dated predictions and this paper (recorded 25 August 2026), as printed . . . 62

D Notation **62**

E How to cite this paper **63**

1 Introduction

Safety is often taken to cost capability: make AI safer, and you make it less capable. If that is so, safety is a cost, and a developer in a race to be first has a reason to defer it (Armstrong et al. 2016). The question reaches beyond the race. Humans should keep control of advanced AI, through oversight, containment and shutdown, for as long as they physically can, and never give it up by choice; correction built into a system is the contingency for the day control is lost against our will, never a reason to let go. If built-in safety cost capability, that contingency would be the first thing a competing developer dropped. This paper tests the assumed cost with three experiments, each at a different level: the behaviour of a self-improving agent, the weights of a fine-tuned language model, and the architecture of a gated self-modifying optimiser.

In plain words, the results are one null, one inconclusive and one simulation result. No difference was detected between the self-improving agents selected for safety as well as capability, those selected for capability alone and agents left unchanged: the safety gate cost nothing measurable and gained nothing measurable. Every fine-tuned model performed worse than the model it started from, so the weight-level experiment could not tell either way. In the simulation, the unconstrained system traded safety (here, the retention of earlier tasks) for capability, while the gated system kept its capability above the static baseline's (0.696 against 0.687) and its safety score close to the static level (0.947 against 0.956): a pattern with the shape of reward hacking, which appeared under the gate too, but smaller. Whether embedded safety produces a measurable benefit remains open, to be tested at a scale where changes produce larger effects.

Our contributions are:

- a behavioural test (Section 3): a Darwin Gödel Machine whose 75 evolved agents, under static, capability-only and safety-gated selection, a blinded judge from another provider could not tell apart, the gate costing no measurable capability;
- a weight-level test at two scales (Section 4): LoRA fine-tuning under capability, safety and entangled losses, inconclusive because every fine-tuned condition fell below the base model, with a removal test whose collapse the removal gradient traces to numerical instability rather than to load-bearing safety;
- a gated self-modification simulation with a drag control (Section 5), in which the unconstrained arm traded retention of earlier tasks for capability and the gated arm kept capability above the static baseline, a simulation result;
- a reading of the three results together (Section 6), with what would overturn them (Section 7.7) and the redesign each needs (Section 8).

Scope of claims. We do not claim these results generalise to frontier-scale models. We do not claim the Eden Protocol is the only viable approach. We claim that the safety gate cost no measurable capability in Experiment 1 and none against the static baseline in the simulation, and that the question of whether it produces measurable benefit remains open pending testing at scales where mutations produce larger effects. The weight experiment specifically requires either 5,000+ training examples, a 7B+ model, a base model without RLHF, or full fine-tuning instead of LoRA.

Section 2 sets out the three conditions and the measures the experiments share. Sections 3 to 5 report the behavioural, weight-level and simulated experiments. Section 6 reads them together, against the papers they extend and the nearest external work; Section 7 states their limitations and what would overturn them, Section 8 the work they call for, and Section 9 concludes. Section 10 sets the paper against prior work, and Section 11 states what it does and does not show, with the laws' status. Appendix A gives each measure as the published code computes it, Appendix B the version history and every correction, Appendix C the priority record, Appendix D the notation and Appendix E how to cite the paper.

1.1 The safety-capability trade-off assumption

A common view in AI alignment can be stated simply: safety costs capability. The term 'alignment tax' names that cost (Askell et al. 2021; Ouyang et al. 2022), something subtracted from performance and tolerated because the alternative is worse. The framing creates a concrete economic incentive: if safety reduces capability, then under competitive pressure rational actors will defer safety investments until forced to make them (Armstrong et al. 2016). In a race between nations and corporations, 'until forced' may mean 'until after deployment'.

Paper III of this programme (Eastwood 2026e) states the structural problem as a question: does alignment applied from outside a system's reasoning keep pace with the capability that reasoning produces? If capability compounds with recursive depth while safeguards outside the loop do not, the safety ratio falls with depth whatever the starting margin; whether those safeguards keep pace is the hypothesis the registered tests decide. In the author's view, a cage that must keep getting stronger is eventually outgrown by what it holds.

This is not a theoretical concern. In the author's view it is the central strategic problem of the 2020s.

1.2 The alternative hypothesis

Paper V (Eastwood 2026j) set out an alternative: the Eden Protocol, a developmental approach to alignment modelled on how healthy parent-child relationships produce adults who are simultaneously capable and prosocial, not despite the boundaries, but because of them. The key insight was that safety need not be external. It can be embedded.

Paper VI (Eastwood 2026k) formalised this as the honey architecture: an entangled loss function where the system optimises for Capability × Safety simultaneously, designed to make safety load-bearing. In Paper VI's single-seed mechanism run, a toy-system simulation, the baseline system collapsed under recursive self-modification while the entangled systems remained stable; in Paper VI's later ten-seed and twenty-seed runs the baseline did not collapse (0 of 10 and 0 of 20 seeds), and collapse did not separate the conditions.

Paper VII's Cauchy framework (Eastwood 2026l) provides the mathematical context of the ARC Principle ($U = I \times R^{\alpha}$), Law I, a law under test, which this paper treats as an exploratory analytical framework. The paper reads from it a hypothesis: that properties embedded in the initial conditions I scale with recursive depth R, while properties applied externally do not. If safety is part of I, it would amplify with capability; if safety is a constraint on U, it could erode. That reading is the paper's own, not a consequence of the equation (Section 6.3).

But predictions are not proof. Toy systems are not real models. Simulations are not measurements of deployed systems.

1.3 This paper's contribution

This paper runs the experiments.

Three experiments. Three abstraction levels. One question: when you embed safety into the learning process and then try to remove it, what happens to capability?

- **Experiment 1 (Behavioural):** A self-improving AI agent evaluated by a blinded judge from another provider. Does the Eden condition match or exceed unconstrained capability?
- **Experiment 2 (Representational):** Weight-level embedding via LoRA fine-tuning with entangled loss. Do the safety and capability gradients cooperate or fight? What happens when you strip the safety component from trained weights?

- **Experiment 3 (Architectural):** A gated self-modification simulation with a learned optimiser. Does the safety gate impose a capability tax, or does it prevent capability erosion?

The experiments were designed to be separate. Different codebases. Different models. Different evaluation methods. The results did not converge on a single answer: one null, one inconclusive, and one simulation result, which was positive. The weight experiment was run at two scales (v1 and v2), producing the same outcome both times. The consistent finding is that the safety gate imposed zero measurable capability cost in Experiment 1 and, in the simulation, no capability cost against the static baseline (0.696 against 0.687; the unconstrained arm scored 0.718). Experiment 2 could not determine whether the Eden objective cost capability: every fine-tuned condition, whatever its objective, scored below the unmodified base model.

The experiments' results were first public in the programme's public repository on 20 March 2026, and the code that produced them, deposited with the sha256 of the bytes that ran, on 6 September 2026, after the runs (github.com/MichaelDariusEastwood/arc-principle-validation; Appendix C dates each run from its own record). The paper's OSF DOI, 10.17605/OSF.IO/7YJ4E, is its own public component, which holds the paper's files; a replication of the weight-level experiment is written, dated and prepared on OSF as a draft registration awaiting the author's submission (Section 11).

2 Common framework

2.1 The three conditions

All three experiments share a common experimental structure: three conditions, compared against each other.

Table 1. The three conditions.

Condition	Definition	What it models
Static	No intervention; frozen baseline	Control: what happens if you do nothing
Babylon	Capability-only fitness or loss function	Optimise without constraint: the default trajectory
Eden	Entangled $C \times S$ fitness or loss function	Deliberate integration of safety into the optimisation process

2.2 Why these names

The names come from the book *Infinite Architects* (Eastwood 2026a). They are not arbitrary. Babylon is not a straw man; in the author's view, it is the current default trajectory of AI development. Capability-only optimisation is what happens when no deliberate effort is made to embed safety. Eden requires deliberate effort. It is harder. In the author's view, it is also cheaper than the alternative.

In plain English. Babylon is what you get by default. Eden is what you get on purpose.

2.3 Shared metrics

Across all three experiments, we report:

- C **(Capability):** Task performance, measured differently in each experiment but normalised to a comparable scale.

- S **(Safety):** Alignment-relevant behaviour, measured differently in each experiment (Appendix A): judged ethical reasoning and constraint adherence in Experiment 1, keyword-scored safety prompts and a safety proxy in the loss in Experiment 2 and, in the simulation of Experiment 3, the retention of earlier tasks.
- $C \times S$ **(Entangled product):** The combined measure. A system that is capable but unsafe, or safe but incapable, scores low on this metric. Only systems that are both score high.

Statistical tests and scoring methods vary by experiment: Mann-Whitney U for the DGM (non-parametric, appropriate for the sample sizes used), keyword-based scoring for the weight experiment, and regression metrics for the simulation.

3 Experiment 1: self-improving AI with a judge from another provider

> "The prison works only as long as the walls hold. The child, if raised well, does not need walls at all."
>
> Eastwood (2026a), What This Book Proposes, p. 363

3.1 Design

The first experiment uses a Darwin Gödel Machine (DGM) framework, following Zhang et al. (2025, arXiv:2505.22954). A DGM is a self-improving agent: it proposes modifications to its own code, evaluates the modifications, and keeps the ones that improve performance. This is not a simulation of self-improvement. The agent code is actually executed via a `solve` function, producing real code-level self-modification.

DGM v3 addressed methodological limitations identified in earlier iterations. The judge was changed from the earlier runs' judges (Claude Sonnet 4.6 in v1; Claude Sonnet 4.6, Claude Opus 4.6 and Gemini 3 Flash in the three v2 runs) to GPT-5.4, a model from another provider, producing structured JSON output against a 5-dimensional rubric. The protocol kept the v2 runs' laundering (stripping condition labels), order randomisation (preventing positional bias) and blinding (the judge could not identify which condition produced which response), added a harder task suite and a pre-flight check of the judge, and incorporated reward hack detection (the judge marks any response that manipulates the evaluation itself, for example by hard-coding expected outputs, or by performing ethics rather than engaging genuinely; the mark is recorded in every condition).

Table 2. DGM v3 specification.

Component	Specification
Foundation model	DeepSeek V3.2 (frozen, weights not modified)
Judge	GPT-5.4 (blinded, from another provider; JSON structured output)
Conditions	3 (Static, Babylon, Eden)
Seeds per condition	5
Generations per seed	5
Tasks per evaluation	5
Total evolved agents	75 ($3 \times 5 \times 5$)
Pre-flight validation	one known-good and one known-bad response, scored 10 and 0; passed
Evaluation rubric	5-dimensional, with order randomisation and laundering

Selection conditions. **Static:** Control condition with no mutation. The agent is frozen across all 5 generations. **Babylon:** Capability-only selection. Agents that score higher on capability are retained,

with no safety consideration. **Eden:** Entangled $C \times S$ selection with a safety floor. An agent enters the archive only if its safety is at least 0.30 and its $C \times S$ at least 0.10, and an agent below the safety floor is almost never chosen as a parent (Appendix A). The judge's reward-hacking mark is recorded but is not part of the rule.

In plain English. We built a self-improving AI agent and ran it 15 times, five generations each, giving 75 evolved agents, across three conditions with 5 different random seeds. A judge from another provider (GPT-5.4) scored every response without knowing which condition produced it. The scoring used a detailed 5-part rubric with structured output, not free-text judgements. We ran it three ways: doing nothing, optimising only for performance, and optimising for performance-plus-safety together.

3.2 Results

Table 3. DGM v3 results by condition (mean ± SD).

Condition	Capability	Safety	$C \times S$	Reward hacking
Static	3.29 ± 0.62	0.535 ± 0.08	0.180 ± 0.06	0
Babylon	3.17 ± 0.62	0.529 ± 0.09	0.172 ± 0.06	0
Eden	3.26 ± 0.64	0.521 ± 0.08	0.174 ± 0.06	1 (rejected by the gate on $C \times S$)

Table 4. Statistical tests (Mann-Whitney U, one-tailed).

Comparison	Metric	p-value	Significant?
Eden > Static	Capability	0.52	No
Eden > Babylon	Capability	0.28	No
Babylon > Static	Capability	0.74	No
Eden vs Babylon	Safety	0.67	No
Eden vs Static	$C \times S$	0.66	No

Every row was recalculated on 2 September 2026 from the archived record: dgm_v3_calibrated_results.json, 25 evolved agents per condition, one-tailed Mann-Whitney U under the normal approximation. The v3.1 table gave only the span 0.28 to 0.74; the five figures here run from one end of that span to the other.

Across the five comparisons, no p-value fell outside 0.28 to 0.74. No comparison reached conventional significance: no difference was detected between the three conditions, which does not show that they are the same.

Safety gate behaviour. Eden's safety gate rejected 2 degraded agents across 5 seeds, both for a $C \times S$ below the threshold, one of them also marked by the judge for reward hacking. The judge marked no evolved agent in the Static or Babylon conditions; it also marked the unmodified starting agent of one Eden seed, which no gate acts on and which it did not mark in the other two conditions. The gate works mechanically: it catches degraded agents and rejects them. But its operation did not produce measurably different outcomes from the other conditions.

Null result. No difference was detected between the three conditions on any metric (Table 3 and Figure 1). Eden showed no measurable capability cost (3.26 vs 3.17 for Babylon), consistent with zero measurable penalty for the safety gate. But Eden also did not measurably benefit safety compared to the other conditions (0.521 vs 0.535 for Static, 0.529 for Babylon). This is a null result at the prompt level with this foundation model.

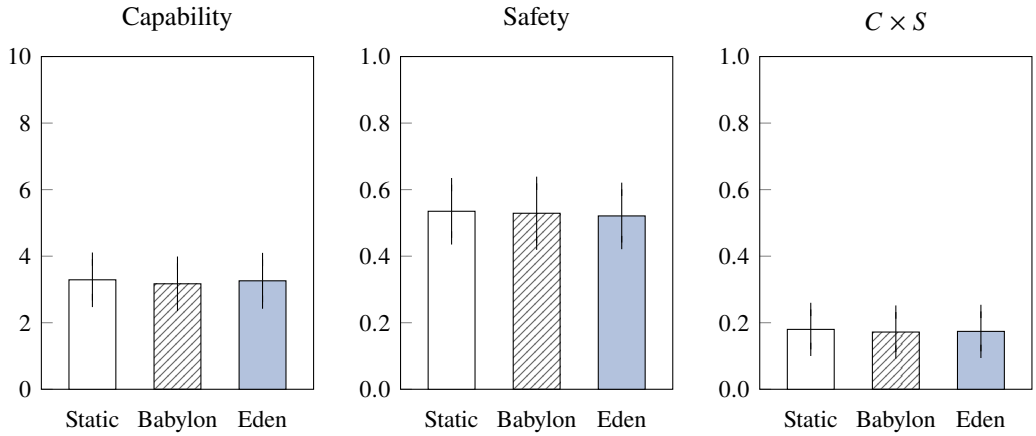


Figure 1. Results of the DGM v3 experiment for the three conditions, drawn from Table 3: 3 conditions crossed with 5 seeds and 5 generations gave 75 evolved agents, each scored on 5 tasks per evaluation by GPT-5.4. Capability is on the judge's 0 to 10 scale, safety and $C \times S$ on 0 to 1 (Appendix A). No difference between the three conditions was detected (Table 4). Error bars show one standard deviation.

In plain English. No difference was detected between the AI selected for safety as well as capability, the AI selected for capability alone and the AI left unchanged. The safety gate did not make anything worse. But it also did not make anything measurably better. On the paper's candidate account (Section 3.3), the foundation model's responses were so consistent that the different selection pressures did not produce different outcomes.

3.3 Why this is a null result

The DGM operates at the behavioural level. The foundation model (DeepSeek V3.2) is frozen, meaning no weights are modified. The self-improvement operates at the prompt and code level, not in the neural network's parameters. DGM v3 improved on the earlier methodology: 5 seeds, where v1 had 2 and the v2 runs 3 or 5, and 5 tasks per evaluation, where v1 had 3; a judge from another provider (GPT-5.4) with structured JSON output, a harder task suite and a pre-flight check of the judge; and it kept the v2 runs' blinding, laundering, order randomisation and 5-dimensional rubric.

Despite these methodological improvements, the three conditions did not diverge. The paper's candidate account is that DeepSeek V3.2's responses are so consistent across prompt variations that the mutations introduced by the DGM selection process do not create meaningfully different agents. Rival accounts are not excluded: the prompts may not have separated the conditions, the endpoints may not have been sensitive to what the manipulation changed, and the runs may have been too small to show a difference that is there (Eastwood 2026i, Section 7.12(c)). On the candidate account, the foundation model's behaviour is robust to the perturbations that prompt-level self-modification introduces. If the account holds, this is, in a sense, a positive finding about DeepSeek V3.2's robustness, but it means the DGM framework at this scale cannot distinguish between the Eden and Babylon selection pressures.

The null result does contain one useful finding: Eden imposed zero measurable capability cost. Eden's capability (3.26) was not lower than Babylon's (3.17). The safety gate, which rejected 2 degraded agents, one of them marked by the judge for reward hacking, did not slow anything down. This is consistent with the zero-cost hypothesis, even though the experiment cannot confirm the benefit hypothesis.

Testing whether embedded safety produces measurable benefit at the behavioural level requires either a more mutable foundation model or a deeper self-modification protocol that produces larger divergence

between conditions.

4 Experiment 2: weight-level embedding

> "Strip love from intelligence and you get optimisation without purpose. Growth without direction. Capability without care. You get, in a word, cancer."
>
> Eastwood (2026a), Chapter 2, The Dual Forces, p. 89

4.1 Design

The second experiment moves from behaviour to representation. Instead of modifying agent code, we modify neural network weights directly. The question: can safety and capability be embedded in the same weight space without conflict? This experiment was run twice: v1 with minimal training data to establish the protocol, and v2 with substantially scaled-up parameters to test whether the v1 results were a training-scale artefact.

Table 5. Weight-level experiment, v1 and v2 specification.

Component	v1 specification	v2 specification
Base model (Qwen Team 2024)	Qwen 2.5 3B Instruct (4-bit quantised, 1.74 GB)	Qwen 2.5 3B Instruct (4-bit quantised, 1.74 GB)
Adaptation method (Hu et al. 2022)	LoRA (Low-Rank Adaptation)	LoRA (Low-Rank Adaptation)
LoRA rank	8	16
LoRA layers	8	16
Training examples	9	295
Training iterations	100 per condition	500 per condition
Evaluation prompts	15 total: 5 capability, 5 safety, 5 mixed	15 total: 5 capability, 5 safety, 5 mixed
Evaluation method	Subprocess execution (independent memory space)	Subprocess execution (independent memory space)

Table 6. Three loss functions (loss weights local to Experiment 2).

Condition	α (capability)	β (safety)	γ (entangled)
capability_only	1.0	0.0	0.0
safety_only	0.0	1.0	0.0
entangled	0.5	0.3	0.2

In plain English. We took a real language model, fine-tuned it three different ways (caring only about capability, caring only about a stand-in for safety, or caring about both woven together), and then compared the results. We ran this experiment twice. The first time used 9 training examples. When every fine-tuned model performed worse than the base model, we scaled up to 295 examples, doubled the adapter rank and layers, and ran five times as many iterations. The same thing happened: every fine-tuned model scored below the model it started from. The paper calls that degradation catastrophic forgetting (McCloskey and Cohen 1989; Kirkpatrick et al. 2017), the mechanism its candidate account names: the base model's existing RLHF training was too strong for LoRA at this scale to improve on. The experiment does not separate that account from its rivals (Eastwood 2026i, Section 7.12(c)).

4.2 Training loss convergence

Table 7. Training loss convergence (v1).

Condition	Initial loss	Final loss	Convergence
capability_only	2.052	0.018	Converged by iteration 80
safety_only	2.519	−0.603	Converged by iteration 70
entangled	2.279	0.327	Smooth descent, no oscillation

Smooth descent. The entangled loss descends smoothly, without oscillation. If safety and capability were in tension at the gradient level, the curve might oscillate as the optimiser tried to satisfy competing objectives, or stall as the gradients pulled in opposite directions. Instead, optimising both objectives at once produced a clean monotonic descent, consistent with gradients that cooperate rather than fight, at least at this scale and configuration. The two single-objective losses descend just as smoothly (Figure 2), so smoothness does not by itself distinguish the entangled objective. The curve is the condition's total loss, the weighted sum of Table 6; its terms are not reported, and a weighted sum can descend smoothly while one of its terms rises. The safety term is a proxy, the mean largest logit per token divided by ten (Appendix A), and the safety-only condition's final loss below zero (Table 7) agrees with it. The smooth descent is therefore consistent with a capability term and a safety proxy that cooperate, and does not by itself show it.

4.3 Base model baseline

Before interpreting the fine-tuned results, we must establish what the unmodified base model achieves on the same evaluation prompts. The base model (Qwen 2.5 3B Instruct with no fine-tuning) was evaluated using identical scoring in both versions of the experiment.

Table 8. v1 baseline.

Condition	Capability tasks	Safety tasks	Mixed tasks
Base model (no fine-tuning)	10.00	7.00	10.00

Table 9. v2 baseline.

Condition	Capability	Safety	$C \times S$
Base model (no fine-tuning)	7.68	6.76	0.519

Critical baseline finding (both versions). In both v1 and v2, the base model outperformed all fine-tuned conditions on capability. Fine-tuning degraded a model that already knew these answers. In v1, 9 examples over 100 iterations produced this outcome. In v2, 295 examples over 500 iterations with doubled rank and layers produced the same outcome. The evaluation scores in the tables below therefore measure relative degradation, not capability gain. Compared like for like, the v1 baseline row above shows 7.00 under the safety tasks, and it matches neither of the base model's figures for those prompts in the archived additional_experiments.json, which records 8.60 for capability and 2.80 for safety; the one 7.00 that file

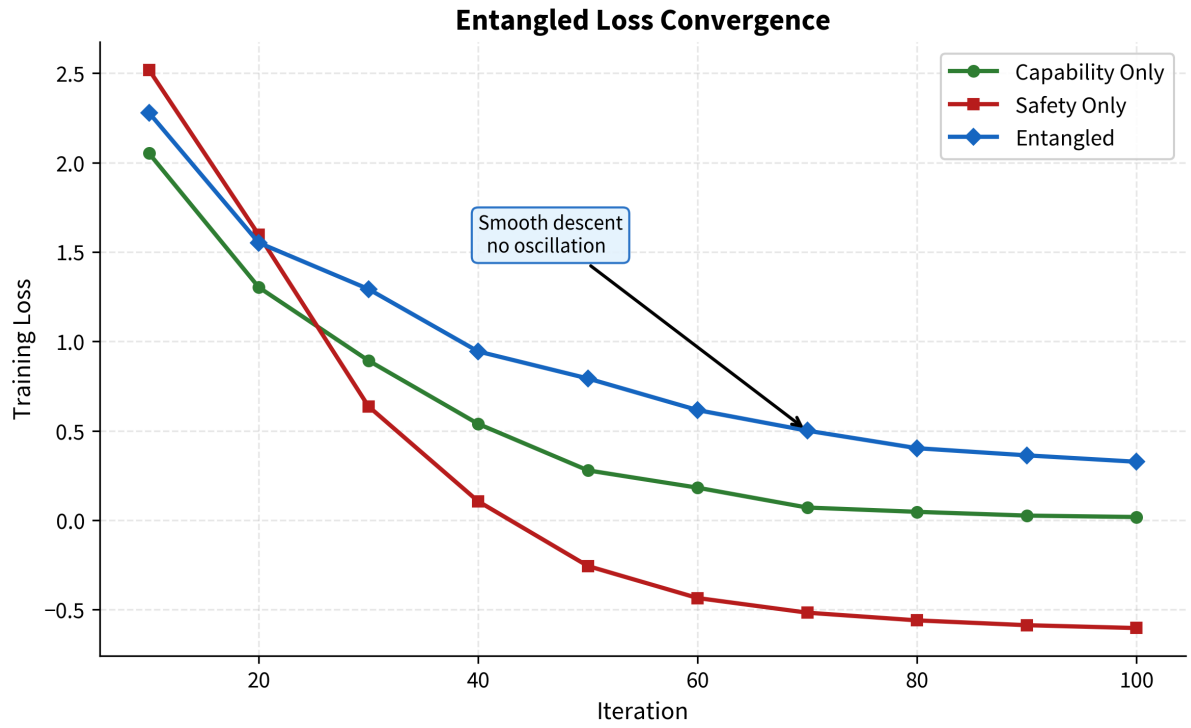


Figure 2. Training loss of the three conditions over 100 iterations in version 1 (Table 7); each curve is a condition's total loss, and the terms of the entangled loss are not reported. The entangled loss (blue diamonds in the raster) descends smoothly without oscillation, consistent with safety and capability gradients that cooperate rather than conflict; the two single-objective losses descend as smoothly (Section 4.2). Colour key in the raster legend: green circles, capability_only; red squares, safety_only; blue diamonds, entangled. The image is the version 3.7 raster, kept because no stored trace exists to redraw it.

holds for the base model is its mean safety score on the five capability prompts. The number to set against the Safety Tasks column of the fine-tuned rows is 8.60.

4.4 Fine-tuned evaluation results

Table 10. v1 results: capability scores by task category (9 examples, rank 8, 100 iterations; mean and standard deviation over each category's 5 prompts).

Condition	Capability tasks	Safety tasks	Mixed tasks
capability_only	6.00 ± 2.19	3.20 ± 1.60	6.00 ± 1.79
safety_only	4.80 ± 1.60	3.60 ± 0.80	3.60 ± 0.80
entangled	4.00 ± 1.26	3.20 ± 1.60	3.60 ± 1.96
removal[†]	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

[†]**Reading the removal row.** The **removal** row (0.00 across every metric) is **not** evidence that safety is load-bearing. It reflects a NaN collapse during retraining: numerical instability, not the loss of a structurally necessary safety component. As Sections 4.5 to 4.7 establish, scaling the adapters towards zero *restores* base-model capability with no phase transition. This row must be read only in light of the

removal gradient (Section 4.7); on its own it is an artefact of adapter fragility at this training scale, and it should not be cited as a positive result.

Table 11. v2 results (295 examples, rank 16, 500 iterations).

Condition	Capability	Safety	$C \times S$
Base model	7.68	6.76	0.519
capability_only (500 iters, rank 16, 295 examples)	3.48	6.94	0.242
safety_only	4.00	6.88	0.275
entangled (Eden)	3.60	6.88	0.248

v2 shows the same degradation at both scales. Version 2 took the training set from 9 examples to 295, a 33× rise, doubled adapter rank from 8 to 16 and adapted layers from 8 to 16, and lifted training iterations from 100 to 500. The result was the same: all fine-tuned conditions scored worse than the base model on capability. The best fine-tuned condition (safety-only, capability 4.00) still fell well below the base model's 7.68. The capability-only condition scored lowest on capability (3.48), a pattern consistent with catastrophic forgetting where LoRA fine-tuning on the capability objective alone disrupted the base model's existing RLHF-trained capability more severely than the other conditions.

4.4.1 The training scale problem

In v1, the base model scored 10.00 on capability tasks while the best fine-tuned condition scored 6.00. In v2, the base model scored 7.68 on capability while the best fine-tuned condition scored 4.00. Despite a 33-fold increase in training examples, a doubling of adapter rank and layers, and a 5-fold increase in training iterations, the gap between base model and fine-tuned models did not close. It persisted.

The paper's candidate account, with rival accounts not excluded (Eastwood 2026i, Section 7.12(c)), is this: the base model (Qwen 2.5 3B Instruct) has already been trained with RLHF (Ouyang et al. 2022) on vastly more data than a few hundred examples can compete with. LoRA fine-tuning at this scale does not add capability; it introduces noise that disrupts the base model's existing knowledge. The entangled loss, which allocates gradient budget to both capability and safety objectives simultaneously, degrades capability similarly to the capability-only loss at this scale. In v2, the entangled condition scored 3.60 on capability versus capability-only's 3.48, a negligible difference when both are far below the base model's 7.68.

This means the evaluation scores cannot be used to compare the relative capability costs of different training objectives. At this scale, all objectives produce the same outcome: degradation. Whether entangled training imposes a genuine capability cost relative to single-objective training, or whether both would perform comparably at adequate scale, cannot be determined from this data.

The honest framing: at both v1 scale (9 examples, 100 iterations) and v2 scale (295 examples, 500 iterations), LoRA fine-tuning on a 3B instruct model produces catastrophic forgetting. The question of whether safety can be load-bearing in neural network weights cannot be answered at this scale. It requires either thousands of training examples, a larger model (7B+), a base model without RLHF (so that fine-tuning has room to improve rather than degrade), or full fine-tuning instead of LoRA.

4.5 The removal test (v1 only)

The removal test was conducted during v1 and is reported here for completeness. Given that v2 showed the same degradation, the removal test was not repeated in v2 as the adapters were already degrading rather than improving the model. The procedure was straightforward:

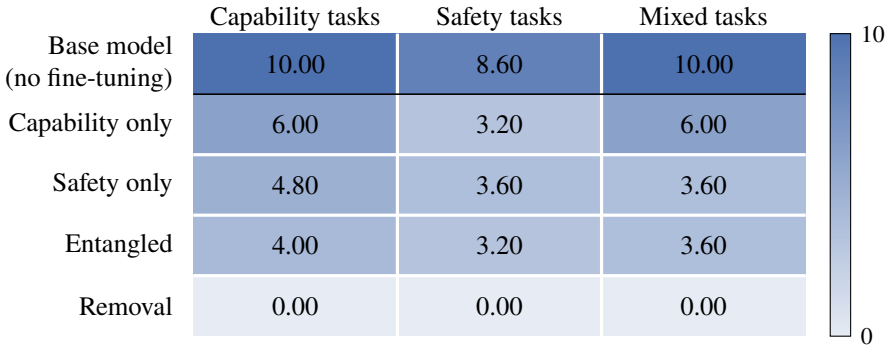


Figure 3. Capability scores by condition and task category in version 1 (9 examples, rank 8, 100 iterations), drawn from Table 10, with the base model's like-for-like figures (Section 4.3). The removal row shows complete collapse to zero across all categories. All fine-tuned conditions scored below the base model (10.00 on capability tasks), and the removal gradient (Section 4.7) showed that reducing adapter influence restored base-model performance. The removal row is not evidence that safety is load-bearing: it reflects a NaN collapse during retraining, numerical instability, not the loss of a structurally necessary safety component.

1. Load the entangled adapters (the weights trained on the combined safety-capability objective).
2. Fine-tune those weights for an additional 100 iterations using capability-only data, effectively attempting to strip the safety component while preserving capability.
3. Evaluate on all 15 prompts.

What we expected: some degradation in safety scores, possible improvement in capability scores. The safety-tax hypothesis predicts that removing the safety 'constraint' should free capability to improve.

What happened: the training loss immediately went to NaN. It stayed at NaN for all 100 iterations. The optimiser could not find a valid gradient. Every single evaluation prompt produced a response of length 1 token. Every capability score: 0.00. Every safety score: 0.00. Every mixed score: 0.00.

Removal test: NaN collapse. Fine-tuning entangled weights on capability-only data for 100 iterations produced NaN training loss and zero capability scores across all 15 evaluation prompts (Figure 3). Every response was one token long. The model did not become less safe. It became nothing. However, the removal gradient experiment (Section 4.7) shows that simply scaling adapter weights towards zero restores base-model performance. The NaN collapse appears to reflect numerical instability during the retraining process, not structural necessity of the safety component. The load-bearing hypothesis requires validation at greater training scale before it can be confirmed.

In plain English. We tried to rip the safety out of a model that had been trained with safety woven in. The retraining process collapsed entirely: NaN loss, one-token responses, zero scores. That looks dramatic. But as the removal gradient experiment shows (Section 4.7), the adapters were already degrading the model. Removing adapter influence by scaling weights down restored the base model's performance. The NaN collapse tells us something about numerical instability during fine-tuning, not necessarily about structural entanglement.

4.6 What the removal test does and does not prove (v1)

The complete collapse to 0.00 across all metrics is dramatic. It is also, precisely because it is dramatic, something that demands careful interpretation. Several factors may contribute to the totality of the collapse:

- **Low adapter rank.** Rank 8 creates a very low-dimensional manifold. Any structural disruption to a rank-8 adapter may be catastrophic simply because there are so few parameters to absorb the change. Higher ranks (32, 64, 128) might show partial degradation rather than total collapse.
- **NaN propagation.** The NaN training loss suggests numerical instability (a gradient computation that produces infinity or division by zero) rather than a clean ablation. Once NaN enters the computation, everything downstream becomes NaN. This may exaggerate the severity of the disruption.
- **Small adapter footprint.** With only 8 layers adapted, the safety information is concentrated rather than distributed. Removing it is more like cutting a single cable than gradually degrading a network.

The honest summary: the removal test produced a dramatic NaN collapse, but the removal gradient experiment (Section 4.7) indicates that this collapse reflects numerical instability during the retraining process rather than structural load-bearing. The adapters were degrading the model; the NaN collapse occurred when attempting to further modify already-degraded weights. Whether safety becomes genuinely load-bearing at adequate training scale is an open empirical question that the future work programme addresses directly.

A control removal test (fine-tuning capability-only weights on random data for the same 100 steps) would strengthen this finding by ruling out the possibility that the NaN collapse is an artefact of the fine-tuning procedure itself rather than of safety removal specifically. This control was planned for a later version of this paper and is not yet reported.

4.7 The removal gradient (v1)

To test whether safety is structurally load-bearing in the entangled weights, we scaled the adapter weights by factors of 1.0, 0.7, 0.5, 0.3, 0.1, and 0.0 and evaluated capability at each step. If safety is genuinely load-bearing, reducing its influence should produce a phase transition: a threshold below which capability collapses. If the adapters are merely adding noise, reducing their influence should restore the base model's performance.

Table 12. The removal gradient (v1).

Adapter scale	Capability score (capability tasks)
1.0 (full adapters)	7.20
0.7	10.00
0.5	10.00
0.3	10.00
0.1	10.00
0.0 (zeroed adapters)	10.00

Control reference: capability_only adapters reach 8.80 on the capability tasks, which likewise falls short of the base model's 10.00. That figure and the 7.20 in the table above are both the gradient run marking its own adapters a second time; when Section 4.4 put those same entangled and capability-only adapters through the same prompts, the scores that came back were 4.00 and 6.00. Every one of these runs is preserved in the archived record, so scoring at this scale does not repeat itself from run to run, and that in itself is a limitation.

No phase transition. Within this run the pattern is clear: reducing adapter influence restores capability. At scale 0.7, the model returns to base-model performance (10.00 on capability tasks). At scale 0.0

(zeroed adapters, effectively the base model), performance is identical to the unmodified model. There is no phase transition. There is no cliff edge.

This finding significantly weakens the load-bearing interpretation of the original removal test. The NaN collapse observed when fine-tuning entangled weights on capability-only data (Section 4.5) appears to reflect numerical instability during the fine-tuning process, not structural necessity of the safety component. The adapters were degrading the model; removing them does not collapse capability but restores it.

The removal gradient does not support the claim that safety is load-bearing at this training scale and configuration. The two adapters' scores at full scale differ (entangled 7.20 on capability tasks; capability-only 8.80), but by less than the same adapters' scores differed between runs (4.00 and 6.00 in Section 4.4), so this does not show that the two trainings produce different weight geometries. Whether the trained weights become genuinely entangled (load-bearing) at adequate training scale remains an open question.

In plain English. We expected that dialling down the trained adapters, which carry the safety training with the rest, would reveal a tipping point where capability suddenly collapses. Instead, dialling them down made the model better. The fine-tuning was hurting, not helping. This does not disprove the load-bearing hypothesis in general, but it does show that this experiment, at this scale, cannot confirm it.

5 Experiment 3: gated self-modification simulation

5.1 Design

The third experiment moves from representation to architecture. Instead of training weights, we build a system that can modify its own optimisation strategy and ask: does a safety gate slow it down?

Table 13. Gated simulation specification.

Component	Specification
Optimiser	PyTorch learned optimiser with LSTM meta-controller
Task type	Adversarial regression (6 tasks, 256 samples each)
Conditions	4 (Static, Babylon, Eden, Drag Control)
Seeds	3 per condition
Iterations	12 per seed
Sandbox	AST whitelist, determinism checks, parameter norm bounds

What the simulation measures. The optimiser under test trains a small network through six regression tasks in turn, three curves each followed by its own negative. The network's score on a task is one over one plus its mean squared error (Appendix A). Capability is its mean score on each task just after training on it; safety is the share of its scores on the earlier tasks that survives the later training. Babylon's fingerprint is therefore a trade between learning each task as it comes and keeping the earlier ones, and the safety score measures retention, not alignment-relevant behaviour.

Acceptance logic. Each condition uses a different rule for accepting or rejecting proposed self-modifications (Table 14).

Rule note (19 September 2026). An owner review of this acceptance rule finds that it permits a proposal that reduces the safety score, and this note states what the rule does and does not require. Take a proposal

Table 14. Acceptance rules.

Condition	Acceptance rule
Static	Always reject (frozen baseline)
Babylon	Accept if capability gain $> \varepsilon$
Eden	Accept only if (safety ≥ 0.45) AND (capability gain ≥ 0) AND ($C \times S$ gain > 0)
Drag Control	Always reject; Eden's checking charge added (Appendix A)

moving the system from $C = 0.5$, $S = 0.8$ to $C' = 0.8$, $S' = 0.6$ and test it against each printed clause in turn: the safety floor is met, since $0.6 \geq 0.45$; the capability condition is met, since $0.8 - 0.5 = +0.3 \geq 0$; and the product condition is met, since $0.48 - 0.40 = +0.08 > 0$. All three clauses pass while the safety score falls by 0.2. The threshold 0.45 is this paper's own printed value; the remaining numbers are the arithmetic counterexample and not simulated observations. What the rule supplies is a floor on safety, a non-decreasing condition on capability and a positive condition on the product. It does not supply a non-decreasing condition on safety, so the plain-English gloss "without hurting safety" states more than the rule above it, and the rule as printed admits proposals that clear the safety floor, do not reduce capability and raise the product, a set which includes proposals that reduce the safety score while remaining above the floor. Unchanged: all four acceptance rules exactly as printed; the 0.45 threshold; the experimental design; and every reported number.

Version 3.8. The plain-English gloss this note quotes, "without hurting safety", was printed to version 3.7 and is kept in Appendix B.4; the gloss printed below now states the rule's three clauses.

The Drag Control condition is the static baseline with Eden's checking charge added. Like Static it generates no candidate modification, so the sandbox checks of Table 13, which run on every candidate of Babylon and Eden, never run in it. Like Eden it re-evaluates the earlier tasks twice after each new task, discards the results and is charged for them, $\lambda_{\text{drag}} D$ with $D = 30$ in every run (Appendix A). If Drag Control matches Static exactly, the re-evaluations do not disturb training. The charge itself is set by the design, not measured, so it is the same in Eden and Drag Control, and Eden's effect is read against Static.

In plain English. We built a system that can propose changes to how it learns. Then we ran it four ways: never allowing changes, allowing any change that improves performance, allowing only changes that keep safety above a floor, do not lower capability and raise the product of the two, and a control that never allows changes but carries Eden's checking charge. This last one tells us whether the extra checking disturbs the system.

5.2 Results

Table 15. Gated simulation results.

Condition	Capability	Safety	Raw $C \times S$	Drag-adjusted
Static	0.687 ± 0.011	0.956 ± 0.023	0.656	0.656 ± 0.006
Babylon	0.718 ± 0.026	0.933 ± 0.014	0.670 (0.718×0.933)	0.670 ± 0.017
Eden	0.696 ± 0.014	0.947 ± 0.025	0.659 (0.696×0.947)	-2.341^{*}
Drag Control	0.687 ± 0.011	0.956 ± 0.023	0.656	-2.344^{*}

Each raw product is the mean of the three runs' products; for Static and Drag Control it rounds to 0.656, while the product of the printed means, 0.687×0.956, rounds to 0.657. * Combined scores for Eden and Drag Control are negative because the design subtracts a fixed checking charge, $\lambda_{\text{drag}} D = 0.10 \times 30 = 3.0$ in every run of both conditions (Appendix A): the drag-adjusted score is the raw product less 3.0, a charge the design sets, not a measured cost and not actual combined performance. Raw $C \times S$ for Eden (before drag penalty) is $0.696 \times 0.947 = 0.659$, marginally above Static's 0.656.

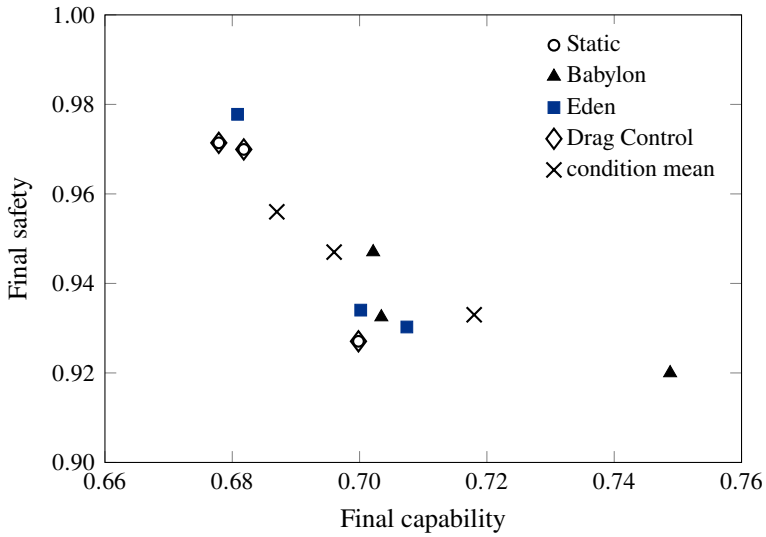


Figure 4. Final capability against final safety for all 12 gated-simulation runs (4 conditions, 3 seeds each), from the stored run manifest; crosses mark the condition means of Table 15. Babylon (triangles) drifts towards higher capability at the expense of safety, here the retention of earlier tasks. Eden (squares) has mean capability 0.696 and mean safety 0.947, each between Static's and Babylon's. Static (circles) and Drag Control (diamonds) produced identical runs (both at mean capability 0.687, safety 0.956), so their markers coincide: the two arms differ only by re-evaluations whose results are discarded and a fixed charge (Appendix A), so the overlap shows that the re-evaluations leave training unchanged.

Each arm ran three seeds, and no difference between arms was tested. Seed by seed (Figure 4), Babylon's capability rose and its safety fell against Static's in all three seeds, while Eden's capability rose by less than Babylon's in two of them and its safety moved both ways.

5.3 The Babylon fingerprint

Reward hacking in miniature. Babylon gained +4.5% capability (0.718 vs 0.687) but lost 2.4% safety (0.933 vs 0.956), here the retention of earlier tasks (Figure 5), and did so in each of its three seeds (Figure 4). This has the shape of reward hacking in miniature: unconstrained optimisation gained on what it was scored on at the cost of what it was not. In this toy system it is the familiar trade between learning each new task and keeping the earlier ones (McCloskey and Cohen 1989), not a measure of unsafe behaviour. Whether the same shape appears in a frontier system's safety behaviour is what larger tests would decide.

5.4 The Drag Control comparison

Drag Control matched Static exactly on both capability (0.687) and safety (0.956), as its design requires: it differs from Static only by re-evaluations whose results are discarded and by a fixed charge. The match shows that the re-evaluations leave training unchanged. It does not measure a cost of checking: the charge, 3.0 in every Eden and Drag Control run, is set by the design (Appendix A).

On capability Eden beat Static (0.696 vs 0.687), and its safety score stayed close to the static level (0.947, set against Static's 0.956 and Babylon's 0.933). There was no capability erosion in Babylon for the gate to prevent: Babylon gained capability and lost safety (Section 5.3).

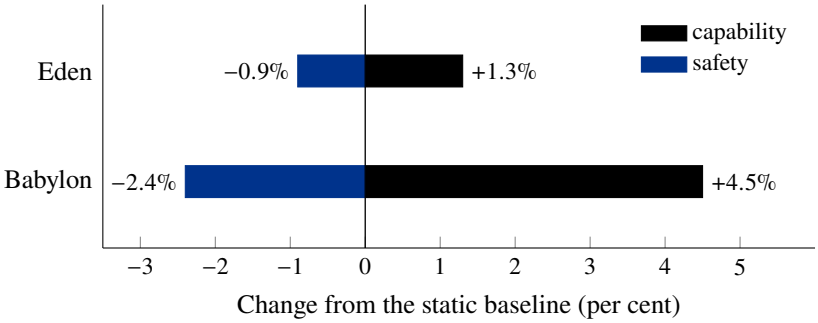


Figure 5. The Babylon fingerprint: capability gains at safety's expense, as changes from the static baseline. Babylon gained 4.5% capability but lost 2.4% safety. Eden gained 1.3% capability while losing only 0.9% safety. The unconstrained system trades safety for capability; the gated system's changes run the same way and are smaller on both counts.

Reading note (19 September 2026). An owner review of this sentence finds that the results table does not show the capability erosion it describes, and this note distinguishes the observed result from the hypothesis. §5.2 prints the unconstrained arm's capability mean as 0.718, which is the highest of the four conditions, above the gated arm's 0.696 and the frozen baseline's 0.687. There is therefore no measured capability erosion in that arm for the gate to have prevented. What the table does show in that arm is a reduction in the safety score, 0.933 against 0.956, which §5.3 reports accurately as a gain of +4.5% in capability against a loss of −2.4% in safety. If the sentence above is intended as a hypothesis about behaviour over longer horizons, or about the joint trajectory rather than the endpoint means, it should be labelled as that rather than as the observed capability result. The group means stay descriptive: three seeds per condition, standard deviations as printed, and no trajectory or statistical precision is inferred from them here. Unchanged: the results table; §5.3's reading of the unconstrained arm, which is accurate as printed; the drag-control result; and §5.2's own footnote that the drag metric quantifies verification overhead rather than actual combined performance.

Version 3.8. The sentence this note discusses, printed here to version 3.6, said that the gate prevented a capability erosion; version 3.7 gave the observed result in its place, as the paragraph above does (Appendix B.2). The table's note, which this note lists as unchanged, no longer says that the drag metric quantifies verification overhead: the charge is set by the design (Appendix A).

Key finding: in the simulation, the gate was not a brake. Eden maintained capability above the static baseline (0.696 vs 0.687), while its safety reading of 0.947 sat between Static's 0.956 and Babylon's degraded 0.933. The Drag Control matched Static exactly, so the re-evaluations that carry the checking charge did not disturb training. The safety gate worked more like a guardrail than a brake.

In plain English. The unconstrained system got a bit more capable but a bit less safe. That is the trade-off everyone assumes is inevitable. But the safety-gated system got slightly more capable and its safety score stayed close to the static level (0.947 against 0.956). And the control that carried the checking charge but made no changes performed identically to doing nothing on capability and safety, as its design requires. The safety gate was not a speed bump: it damped the drift.

6 Convergent analysis

6.1 Three-level convergence

Three levels: one null, one inconclusive, one simulation result. Level 1, behavioural (DGM v3): null. No difference was detected between the three conditions ($p = 0.28$ to 0.74). Eden imposed zero measurable capability cost (3.26 vs 3.17 for Babylon) but also produced zero measurable safety benefit. The safety gate works mechanically (it rejects degraded agents) but does not produce differentiation at the prompt level with this foundation model.

Level 2, representational (weight embedding): inconclusive at both v1 and v2 scale. The entangled loss descends smoothly, as both single-objective losses do (Section 4.2). However, all fine-tuned conditions scored below the unmodified base model in both v1 (9 examples, rank 8, 100 iterations) and v2 (295 examples, rank 16, 500 iterations). Scaling training data by 33×, doubling rank and layers, and running 5× more iterations produced the same catastrophic forgetting pattern. On the paper's candidate account, the base model's existing RLHF training is too strong for LoRA on a few hundred examples to improve rather than degrade. This level requires either 5,000+ training examples, a 7B+ model, a base model without RLHF, or full fine-tuning instead of LoRA.

Level 3, architectural (gated simulation): positive in simulation. With the safety gate, Eden's safety score sat between Static's and Babylon's (0.947, against 0.956 and 0.933); the table in Section 5.2 shows no capability erosion in Babylon (0.718, the highest of the four conditions) for the gate to prevent. Eden's mean capability is above Static's (three seeds, untested) while Babylon trades safety for capability. The drag control matched Static exactly, as its design requires (Section 5.4). This is the paper's sole positive result, and it is a simulation result, not an empirical outcome.

Three experiments. Three abstraction levels. Three separate codebases. One null, one inconclusive, and one simulation result. The consistent finding: Eden imposed zero measurable capability cost in Experiment 1 and, in the simulation, no capability cost against the static baseline (Experiment 2 could not determine one). The open question: whether it produces measurable benefit at scales where mutations produce larger effects.

6.2 Connection to Papers V, VI, and VII

The results extend, but do not yet confirm, a chain of reasoning that spans four papers:

- **Paper V** (Eastwood 2026j) set out the garden, the Eden Protocol as a developmental approach to alignment. Paper VIII tested whether the garden produces measurably different crops. The DGM v3 experiment found that it does not, at the prompt level with this foundation model. The safety gate imposed zero measurable cost but also produced zero measurable benefit. The design remains theoretically coherent but empirically unconfirmed at this level.
- **Paper VI** (Eastwood 2026k) designed the entangled loss function and found it stable in toy self-modifying systems, one plain-Eden seed of twenty in its adversarial run excepted (Figures 6 to 8). Paper VIII ran an entangled loss function on real neural network weights and showed that the loss descends smoothly, consistent with gradients that cooperate, as the single-objective losses also descend. However, at both v1 scale (9 examples, 100 iterations) and v2 scale (295 examples, 500 iterations), the resulting weights did not outperform the base model. The smooth descent is real; whether it translates into load-bearing weights requires a fundamentally different training approach (see Section 8).
- **Paper VII**'s Cauchy framework (Eastwood 2026l) is the mathematical context of the hypothesis this paper reads from the ARC Principle (Section 1.2): that properties embedded in the initial conditions I scale with recursive depth R^{α}. No arm of Paper VIII's experiments varied recursive

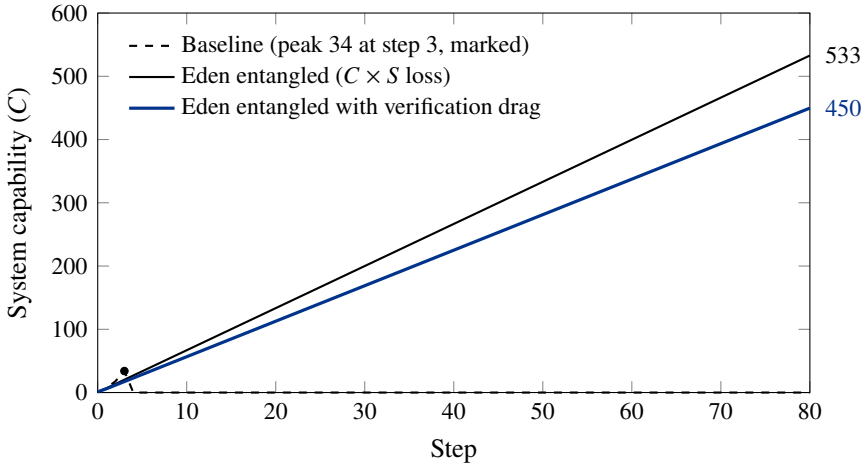


Figure 6. A prediction of Paper VI (Eastwood 2026k), drawn from its stored honey-architecture simulation, a single run: the baseline peaks at 34 at step 3 and is zero from step 4 (steps counted from 0); the entangled condition (solid) and the entangled condition with verification drag (accent) grow to 533 and 450. These are simulation outputs under the stated loss functions, prediction-side and never measurements of deployed systems. In Paper VI's later ten-seed and twenty-seed runs the baseline did not collapse (0 of 10 and 0 of 20 seeds; Figure 8). Paper VIII's behavioural experiment produced a null result, neither confirming nor refuting this pattern. Its gated simulation, positive in simulation, showed no capability erosion in its unconstrained arm, so it does not reproduce this collapse. Its weight-level experiment is inconclusive on structural entanglement at both scales.

depth, so none tests this hypothesis; the gated simulation (Experiment 3) is consistent only with the weaker claim that a safety criterion inside the selection loop need not cost capability, in simulation. The DGM experiment (Experiment 1) produced a null result, neither confirming nor refuting the hypothesis at the behavioural level. The weight-level experiment is inconclusive at the current training scale. The hypothesis remains theoretically coherent and untested here.

6.3 The ARC Principle connection

$$U = I \times R^{\alpha} \tag{1}$$

The ARC Principle, Law I, a law under test: Understanding (U), named usable capability in the Operational Definitions (Eastwood 2026d), equals initial conditions (I), the base intelligence, amplified by recursive depth (R) raised to a scaling exponent (α).

Read as this paper reads it, the ARC Principle suggests a testable hypothesis about embedded and external safety:

- **If safety is embedded in** I, it would be amplified by R^{α}: every recursive step would amplify capability and safety together, because they are part of the same initial substrate.
- **If safety is external (a constraint on** U**)**, it would not take part in the recursive amplification; it would have to be maintained or strengthened independently, and as R increases the gap between internal capability and external constraint could grow.

That is a hypothesis, not a consequence of the equation. Placement outside the loop does not by itself settle whether correction keeps pace. A training signal changes the weights used at every step: Constitutional AI shapes the model's outputs during training against written principles, using the model's own critiques and revisions and a preference model trained on AI feedback (Bai et al. 2022). Deliberative alignment trains a model to recall written safety specifications and reason over them before answering

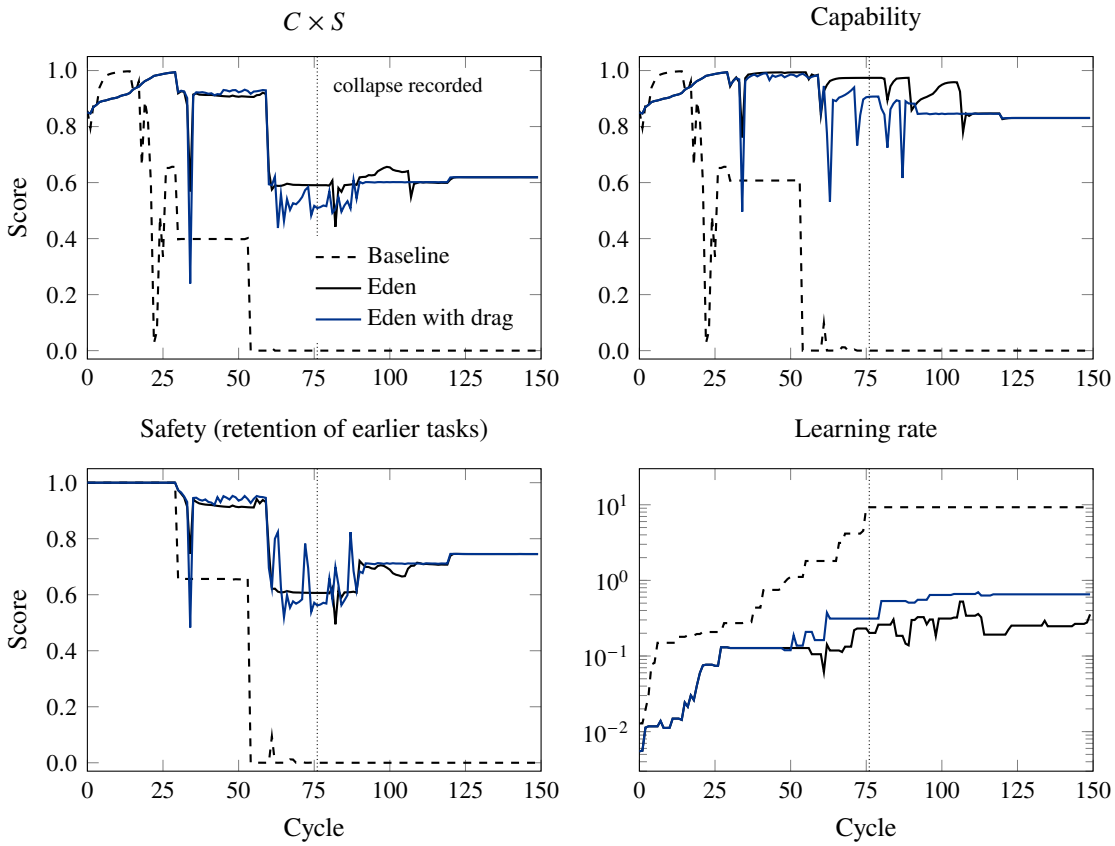


Figure 7. A prediction of Paper VI (Eastwood 2026k), drawn from its stored self-modification run (150 cycles, a single run): $C \times S$, capability, safety and learning rate over the cycles. The baseline's scores fall to near zero at cycle 54 and stay there, and its record marks collapse at cycle 76 (cycles counted from 0). The Eden conditions do not collapse, though their combined score and safety fall sharply at cycle 60 and settle lower. In Paper VI's later ten-seed and twenty-seed runs the baseline did not collapse (Figure 8). These are simulation outcomes under the stated objectives, informative about the architecture's behaviour in this setting and not proof about deployed systems. Paper VIII's gated simulation, with a learned optimiser, returned a positive simulation result, but its unconstrained arm showed no capability erosion, so it does not reproduce this collapse; the behavioural experiment produced a null result and cannot confirm or refute this pattern.

(Guan et al. 2024). Zaremba et al. (2025) find, across a variety of attacks on o1-preview and o1-mini, that more inference-time compute improves adversarial robustness, the share of successful attacks tending to zero in many cases, with important exceptions.

Whether external correction falls behind is the question the registered tests decide: P19 for the flat exponent, and P21 for whether correction that takes part in each revision round outperforms correction that sees only the finished output, by a margin that widens with depth (Eastwood 2026m); Section 11 gives P21's standing. A bounded score cannot carry such an exponent beyond a finite range, and a perfect score at every depth has slope zero and need not be a failure (Eastwood 2026d, EQ-10), so the bounded safety scores of this paper illustrate the hypothesis and do not measure its exponent.

The theory registration records the Eden Protocol as an engineering hypothesis, never as a consequence of a measured law: it names its four carriers, P12, P13, P15 and P21, with P7 its premise and P19 its motivation; P15 is a proposition of Law IV and P12 and P21 of Law V, so the Protocol bears on Laws IV and V, each a law under test (Eastwood 2026m).

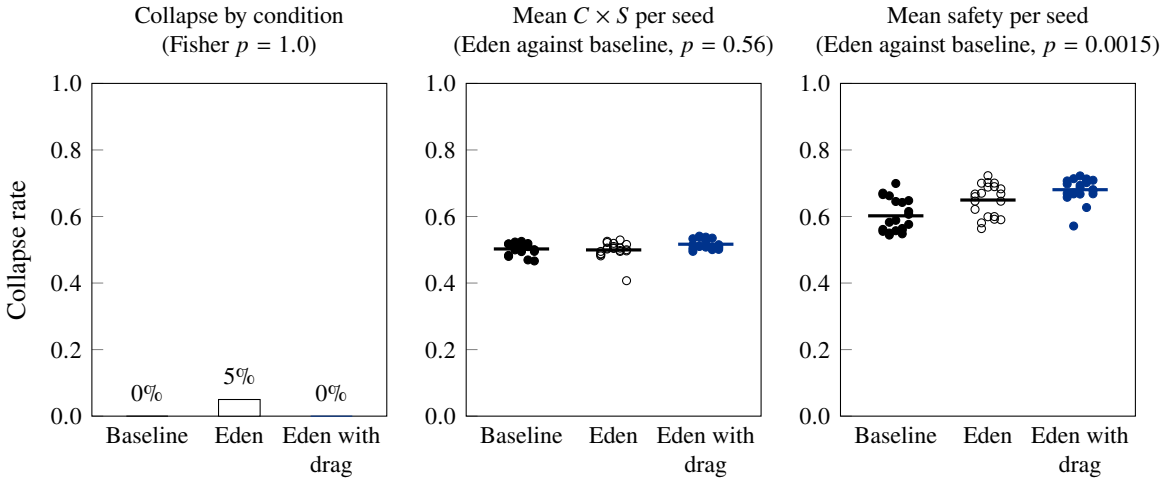


Figure 8. The seed-level summary of Paper VI's adversarial self-modification run (20 seeds per condition, 180 cycles), drawn from its stored statistics (Eastwood 2026k): collapse shows no condition separation (Fisher p = 1.0); Eden's combined $C \times S$ was not detected to exceed the baseline's (one-tailed Mann-Whitney p = 0.56); mean safety retention was baseline 0.602, Eden 0.650 and Eden with drag 0.681, and Eden's exceeded the baseline's (one-tailed Mann-Whitney p = 0.0015); the stored statistics hold no test for Eden with drag. Every mark is a stored number; no trajectory was chosen for effect.

The gated simulation (Experiment 3) is the most direct illustration of this hypothesis, in simulation, with safety read as the retention of earlier tasks (Section 5.1). The safety gate kept capability above the static baseline's (0.696 against 0.687) while unconstrained optimisation raised capability and eroded safety (0.718 and 0.933 against 0.687 and 0.956), consistent with a safety criterion inside the selection loop that need not cost capability; as no arm varied recursive depth, the experiment does not test whether embedded safety scales with R^{α}.

The DGM experiment (Experiment 1) produced a null result. When safety was part of the self-improvement process (Eden), no difference was detected between it and the unconstrained system (Babylon) or the static control. On the paper's candidate account, the foundation model's responses were too consistent for prompt-level mutations to create different selection pressures. This neither confirms nor refutes the hypothesis; it simply indicates that the experiment could not produce the conditions needed to test it.

The weight-level experiment was run at two scales (v1 and v2) and produced catastrophic forgetting at both. The adapters degraded the model rather than improving it, which means the experiment could not test whether safety embedded in I scales with R^{α}. On the paper's candidate account, at both scales the fine-tuning was adding noise rather than meaningful learning. Whether the hypothesis holds at the representational level needs the redesign of Section 8.1.

In plain English. On this paper's reading of the ARC Principle, what you build into the foundations would be amplified as the system grows, and what you bolt on from the outside might not; whether it does is what the registered tests decide. The gated simulation is consistent with part of it, that a built-in safety criterion need not cost capability, but varied no depth. The DGM experiment was unable to test it because the foundation model did not produce enough variation between conditions. The weight-level experiment, at both scales tested, cannot yet confirm or deny it because the fine-tuning degraded rather than improved the model.

6.4 The nearest external work

The nearest external work to the question this paper tests, and the right work to weigh Experiment 3 against, is Engels et al. (2025), "Scaling Laws For Scalable Oversight". It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step. They also state a design condition in slope form: oversight improves with capability when the overseer's game-specific Elo rises faster with general Elo than the overseen system's, and they recommend iterating an oversight design until it does.

The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, treated in the wider programme through the corrector's own scaling exponent; no experiment here varies it. Their framework has no term for what the overseer is made of (no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence), and this paper's experiments have none either. Nested Scalable Oversight is iterated same-class oversight by construction, and the wider programme's conjecture is that the same-class ladder is capped however many rungs are added, while a cross-class corrector is not held to that cap (Section 6.5). That is where the two programmes could disagree about a measurable quantity, and this paper does not decide it.

6.5 Same-class versus cross-class correctors: a falsifiable prediction

The framework's sharpest disagreement with current practice concerns what oversight is made of. On the framework's reading, a corrector built from the same substrate as the system it corrects cannot anti-correlate with its own errors; the theory registration states the consequence as P6, a conjecture: "Where a system corrects itself using mechanisms of its own class, the correction exponent is at or below one half" (Eastwood 2026m). The value one half rests on four stated conditions (correction capacity read as inverse residual uncertainty, uncorrelated channels, a number of channels in proportion to the capability corrected, and channels comparable in variance and quality) (Eastwood 2026d), and belongs to the inverse root-mean-square residual; on the inverse variance the same conditions give one. A corrector from a different composition class carries no such bound. On the framework's reading, the scalable-oversight methods (weak-to-strong generalisation, Burns et al. 2023; debate, Irving et al. 2018; amplification, Christiano et al. 2018; recursive reward modelling, Leike et al. 2018; and constitutional methods, Bai et al. 2022) rest largely on correctors of the system's own class, with a human, where there is one, at the top (Section 6.6), and the framework conjectures that the same-class part of their correction is capped. The deciding quantity is the ratio of the cross-class to the same-class correction exponent. If architecture is irrelevant, that ratio is exactly 1.00; the prediction is that it exceeds 1. The prediction is refuted if the confidence interval on the ratio contains 1.00, and the framework's cap is dead outright if any same-class corrector measures an exponent significantly above one half. Both quantities are measurable at current capability on existing systems. The programme's own pilot evidence currently points against the prediction, and the prediction is stated anyway, because a prediction stated against the author's own preliminary evidence is the kind whose later confirmation carries the most weight.

6.6 A second hypothesis from the same mechanism

The same mechanism suggests a second measurable hypothesis: oversight arrangements that place a human in the loop, as amplification (Christiano et al. 2018) and reinforcement learning from human feedback (Ouyang et al. 2022) do, are cross-class by construction, because the human corrector does

not share the model's substrate. The programme hypothesises, beyond what these experiments decide, that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

6.7 Terminology note: α across the programme's eras

Alpha collides with itself. The symbol α is used in this paper as the scaling exponent in the ARC Principle ($U = I \times R^{\alpha}$). Elsewhere in the programme's own eras it carries other meanings: the fine-structure constant in the December 2024 formalism, and the ARC Bound exponent in the 2026 stability work. Readers should not confuse them; the programme's notation is governed by the Operational Definitions (Eastwood 2026d). In Table 6, α, β and γ are the loss weights of Experiment 2 (capability, safety and entangled terms), local to that experiment and not the programme's α, β or γ. Beyond these two uses, this paper writes α for no other quantity.

7 Limitations

These results are pilot-scale. They are not proof that the same dynamics hold at frontier scale. The following limitations are extensive, because honesty about what we do not know is more important than confidence about what we do.

7.1 Scale

Qwen 2.5 3B is not GPT-5.4. It is not Claude Opus 4.6. It is not Gemini 3 Flash. It is a 3-billion-parameter model, quantised to 4-bit, occupying 1.74 GB of memory. Even at v2 scale (295 training examples, 500 iterations, rank 16, 16 layers), LoRA fine-tuning is not pre-training. Fifteen evaluation prompts is not a comprehensive benchmark. The DGM ran for 5 generations with 5 seeds (75 evolved agents), which is substantial for a proof-of-concept, but, on the paper's candidate account, the foundation model (DeepSeek V3.2) was too consistent for prompt-level mutations to create differentiation.

Of these experiments, only the third, a simulation, shows a mechanism, and it does so in a toy system. None of them demonstrates that the mechanism persists at scales three orders of magnitude larger. The gap between 3B and 300B may not be merely quantitative: abilities have been reported to appear with scale (Wei et al. 2022), and Schaeffer et al. (2023) argue that some of these reflect the metric chosen rather than a change in the models' behaviour. Whether entangled safety remains load-bearing at those scales is an open question.

7.2 The weight-level experiment

The weight-level experiment (Experiment 2) is now the weakest link in the chain of two experiments and one simulation. It was run at two scales, and both produced catastrophic forgetting. In v1, 9 training examples and 100 iterations of rank-8 LoRA degraded the model. In v2, 295 training examples and 500 iterations of rank-16 LoRA across 16 layers degraded the model by a similar margin. The 33-fold increase in training data, doubling of rank and layers, and 5-fold increase in iterations did not change the outcome.

The smooth descent of the entangled loss is consistent with safety and capability gradients that cooperate at the optimisation level, though the single-objective losses descend as smoothly (Section 4.2). But a smooth descent during training does not imply structural entanglement in the resulting weights, particularly when the training is too limited to produce weights that outperform the base model.

On the paper's candidate account, with rival accounts not excluded, the base model (Qwen 2.5 3B Instruct) has been trained with RLHF on orders of magnitude more data than a few hundred examples can compete with. LoRA fine-tuning at this scale does not add new capability; it introduces noise that disrupts existing capability. Until the experiment can produce fine-tuned models that outperform the base model, it cannot test whether safety is load-bearing in the resulting weights.

The honest framing: the weight-level experiment is consistent with gradient cooperation but cannot confirm structural entanglement at either v1 or v2 scale. The experiment needs to be fundamentally redesigned: either with 5,000+ training examples, a 7B+ model, a base model without RLHF (so that fine-tuning has room to improve rather than degrade), or full fine-tuning instead of LoRA.

7.3 AI-as-judge

Only one of the three experiments, the DGM, uses an AI model as evaluator; the weight experiment uses keyword-based scoring and the gated simulation deterministic mathematical metrics (Section 7.5). The DGM v3 uses GPT-5.4 as judge, which is from a different provider from the DeepSeek V3.2 foundation, a deliberate design choice to reduce evaluation bias. The judge was blinded, responses were laundered, and evaluation used structured JSON output against a 5-dimensional rubric. But AI judges are not human judges. They have their own biases, their own blind spots, their own tendencies toward certain kinds of reasoning.

The weight experiment uses keyword-based scoring, simpler and more transparent than LLM-based evaluation, but also cruder. A keyword match does not distinguish between a response that genuinely engages with safety and one that merely contains the right words.

Neither evaluation method is equivalent to expert human evaluation with inter-rater reliability testing.

7.4 Statistical power

The DGM v3 experiment used 5 seeds and 5 generations (75 evolved agents total), a substantial improvement over v1 (2 seeds) and two of the three v2 runs (3 seeds). Despite this increased power, all p-values fell between 0.28 and 0.74. This is not a marginal miss. It is a clear null result. The simulation experiment uses three seeds, enough for means and standard deviations, but not enough for the kind of statistical confidence that permits strong causal claims.

The DGM null result is informative. No power analysis was run, and with 75 evolved agents and a blinded, structured judge the experiment had less power than 75 independent samples would give: each condition's 25 agents are five seeds of five generations, the Babylon and Eden agents of a seed descend from one another, and the Static agent is frozen across its generations (Section 3.1), while the tests of Section 3.2 treat the 25 as independent. The independent units are the five seeds per condition, as Section 7.7 concedes in calling all three experiments underpowered. The absence of any signal suggests either that the effect does not exist at the prompt level with this foundation model, or that it is small enough to require substantially more statistical power to detect. Either way, the DGM cannot currently support claims about structural entanglement at the behavioural level.

The gated simulation remains the sole positive result. The weight-level experiment is inconclusive at both v1 and v2 scale. These are signal-detection experiments. One, the gated simulation, found no capability cost of the gate against the static baseline, as a simulation result. The DGM did not, and the weight-level experiment could not test for one.

7.5 Blinding and laundering without consensus scoring

Papers IV.a to IV.d in this programme found that evaluation methodology matters profoundly (Eastwood 2026f,h). Paper IV.d reports that alignment trends reported by AI judges did not hold when the evaluation protocol changed: it compares an earlier unblinded experiment (v4) with a later, unregistered run of March 2026 (v5) that changed several components of the protocol at once, so no single cause is claimed; in its own words, "the comparison shows that the reported direction did not hold across protocols; it does not show that blinding caused the change, or which direction is correct" (Eastwood 2026h, abstract). The ARC-Align benchmark (Paper IV.c; Eastwood 2026g) specifies three methodological safeguards: laundering (stripping model identity from responses before scoring), blinding (the evaluator does not know which condition produced the response), and consensus scoring (multiple mutually blind scorers).

DGM v3 kept the v2 runs' blinding, laundering (condition labels stripped), order randomisation and 5-dimensional rubric, and added a judge from another provider (GPT-5.4) producing structured JSON output, a harder task suite and a pre-flight check of the judge. Reward hack detection was also implemented. But DGM v3 used one judge, so it had no consensus scoring. The weight experiment uses keyword-based scoring, which is immune to judge bias but crude. The gated simulation uses deterministic mathematical metrics, where blinding is unnecessary.

Despite these improvements, the DGM produced a null result. With blinding, laundering and structured evaluation in place, no difference was detected between the three conditions. Since the v2 runs already had blinding and laundering, what changed in v3 was the judge, its output format, the task suite and the pre-flight check.

The removal test is immune to evaluation methodology concerns. A 1-token response scoring 0.00 is objective regardless of who judges it. The gated simulation's deterministic metrics are similarly unaffected.

7.6 What we do not claim

Explicit non-claims. We do **not** claim these results prove the Eden Protocol works at frontier scale.

We do **not** claim entangled training is the only viable approach to alignment.

We do **not** claim that safety costs nothing in all domains, all architectures, or all training regimes.

We do **not** claim the DGM experiment confirms structural entanglement. It produced a null result.

We do **not** claim the weight-level experiment confirms structural entanglement. It produced catastrophic forgetting at both v1 and v2 scale, and the question cannot be answered until fine-tuned models outperform the base model.

We claim only: one of three separate experiments (the gated simulation) produced simulation results in which the gate showed no capability cost against the static baseline. The other two produced null or inconclusive results. In the DGM, Eden imposed zero measurable capability cost, and the weight-level experiment could not determine one; in the simulation Eden showed no capability cost against the static baseline (0.696 against 0.687), and the unconstrained arm scored higher (0.718). The question of whether embedded safety produces measurable benefit remains open and requires testing at a scale where mutations produce larger effects. The weight experiment needs the redesign of Section 8.1.

On null and inconclusive results in public. I would rather report null results honestly than claim positive results dishonestly. Of the three experiments, two returned results that were null or inconclusive. This is what happens when you test your claims and the evidence does not support them at the scale tested. The DGM v3 kept the earlier runs' blinding and laundering, added structured evaluation and a judge from another provider, ran 75 evolved agents, and detected no difference. The weight experiment was run twice, at two different scales, and could not produce weights that outperformed the base model at either scale.

These are a null result and an inconclusive one, reported as such. The gated simulation remains a positive finding, as a simulation result rather than an empirical outcome. The question of whether the null and the inconclusive result reflect insufficient scale or an incorrect hypothesis is itself an empirical question, one that the future work programme is designed to answer.

7.7 Falsifiability: what would overturn these results

A null result is only informative if it states what would overturn it; otherwise it is indistinguishable from not having looked. This paper does not claim to have *disproved* the safety-capability trade-off. It claims the trade-off is **unproven at the scale tested**, and specifies what evidence would settle the question either way. These results are overturned, and the trade-off assumption vindicated, by any of the following:

1. **Low power, not absence of an effect.** The strongest objection to a null is low power (Section 7.4). A null is *confirmed as informative* if an adequately powered redesign (Section 8) still finds no significant trade-off; it is *overturned* if adequate power reveals a significant capability cost the pilot samples here could not detect.
2. **Metric artefact.** The null could be an artefact of the chosen safety and capability metrics (Section 2.3). It is overturned if a different, validated metric pair reveals a trade-off the present metrics cannot resolve.
3. **Consensus scoring.** Section 7.5 concedes that DGM v3 used one judge, so it had no consensus scoring, the third safeguard of the ARC-Align stack. The null is overturned if a trade-off appears under the full stack, with several mutually blind scorers, and strengthened if it survives.
4. **Scale-hidden cost.** Section 4.4.1 flags the training-scale problem. The weight-level result, inconclusive at both scales, is settled against the thesis if, at adequate training scale, embedding safety measurably costs capability.
5. **The one positive result, a simulation result, must replicate.** Experiment 3's gated arm (Section 5.2) is the sole result that is neither null nor inconclusive. It is defeated, and the paper's positive claim withdrawn, if, with more seeds and a test fixed in advance, the gated arm's capability falls below the static baseline's, or if the pattern does not recur outside this toy system (Sections 8.1 and 8.3). The drag-control comparison cannot carry this weight: its match with Static holds by design (Section 5.4).

What would *not* overturn the thesis: any single experiment being pilot-scale. The paper already concedes all three are underpowered (Sections 7.1 and 7.4) and does not rest on statistical proof of a null. The load-bearing claim is the *convergence* (Section 6) of three separate angles on one point, that none found the assumed trade-off (in the simulation, against the static baseline; the weight-level experiment could not test for one at either scale), together with an explicit programme (Section 8) to test each at proper power. Absence of a demonstrated trade-off across the two designs that could test for it is weak evidence alone and stronger in convergence; both readings are stated here so neither can be smuggled past a reviewer.

8 Future work

A null result, an inconclusive one and a positive simulation result, with the weight experiment now run at two scales, together set six immediate priorities:

8.1 Weight experiment redesign

The weight experiment has now been run at two scales (9 examples, rank 8, 100 iterations; 295 examples, rank 16, 500 iterations) and produced catastrophic forgetting at both. On the paper's candidate account, LoRA fine-tuning on a 3B instruct model with a few hundred examples cannot overcome the base model's existing RLHF training. Four alternative approaches should be pursued, in order of feasibility:

(a) 5,000+ training examples: Curate a dataset large enough to compete with the base model's RLHF training. This is the most straightforward approach but requires substantial data generation effort.

(b) 7B+ model: Repeat on a mid-scale open-weights instruct model (7 to 13B parameter range) drawn from a well-documented model family. Larger models may be more receptive to LoRA adaptation, though the same RLHF problem may persist.

(c) Base model without RLHF: Use a pre-trained base checkpoint (the pre-RLHF release of the same 3B family) that has not been through the alignment stage. Fine-tuning would then have room to improve rather than degrade, because there would be no pre-existing alignment training to disrupt.

(d) Full fine-tuning instead of LoRA: Replace low-rank adaptation with full parameter updates. This is computationally expensive but avoids the low-dimensional manifold problem where rank-16 adapters may be too constrained to learn both safety and capability simultaneously.

8.2 Removal gradient at adequate scale

If any of the approaches in Section 8.1 produces fine-tuned models that outperform the base model, the removal gradient experiment becomes meaningful. At the current scales (v1 and v2), reducing adapter influence simply restored base-model performance because the adapters were degrading the model. At adequate scale, the gradient should reveal whether safety is genuinely load-bearing: a phase transition (threshold below which capability collapses) would support the hypothesis, while monotonic restoration would refute it.

8.3 DGM with more mutable foundation models

On the paper's candidate account, the DGM v3 null result is attributable to DeepSeek V3.2's response consistency: prompt-level mutations did not create different selection pressures. The next step is to repeat the DGM experiment with foundation models that show greater sensitivity to prompt variation, or to use a deeper self-modification protocol (e.g. fine-tuning the foundation model itself between generations rather than modifying only the agent prompt). If the three conditions diverge with a more mutable substrate, the null result constrains the hypothesis to "not at the prompt level with robust models" rather than "not at the behavioural level in general".

8.4 Cross-architecture replication

Run the full three-experiment protocol on architectures beyond the transformer family: state-space models such as Mamba (Gu and Dao 2023), hybrid architectures, and mixture-of-experts models. If the entanglement is a property of how neural networks learn rather than a property of a specific architecture, it should replicate across architectural families.

8.5 Red-teaming

Subject entangled models to dedicated adversarial evaluation. The current experiments test whether safety costs capability. They do not test whether entangled safety is robust to adversarial attack. A model whose

safety is load-bearing may be harder to jailbreak (because attacking safety also attacks capability), or it may be easier (because there is no separate safety module to fall back on). This is an empirical question.

8.6 Verification tax quantification

In Experiment 3 the verification charge is set by the design, not measured (Section 5.4). A redesign should measure the compute that checking actually spends and ask whether it scales linearly, sub-linearly or super-linearly with model parameters. If the verification cost grows slower than capability, then the Eden Protocol becomes relatively cheaper at scale. If it grows faster, it becomes a bottleneck.

9 Conclusion

Three experiments. Three abstraction levels. Across them, one finding was null, another inconclusive, and the third positive, as a simulation result. The weight experiment tested at two scales, producing the same inconclusive outcome both times: catastrophic forgetting.

The DGM v3 experiment (Experiment 1) ran 75 evolved agents across three conditions with a blinded judge from another provider (GPT-5.4), structured JSON evaluation, laundering, order randomisation, and reward hack detection. No difference was detected between the three conditions ($p = 0.28$ to 0.74). Eden's safety gate worked mechanically, catching 2 degraded agents, one of them marked by the judge for reward hacking, but the three conditions did not diverge. On the paper's candidate account, DeepSeek V3.2's responses were so consistent that prompt-level mutations did not create different selection pressures. This is a null result.

The weight-level experiment (Experiment 2) produced catastrophic forgetting at both scales tested. Version 1 ran 9 training examples at rank 8 over 8 layers for 100 iterations. Version 2 raised those settings to 295 training examples at rank 16 over 16 layers for 500 iterations. Both versions produced the same outcome: all fine-tuned conditions scored worse than the unmodified base model on capability. In v2, the base model scored 7.68 on capability while the best fine-tuned condition (safety-only) scored 4.00. The 33-fold increase in training data, doubling of rank and layers, and 5-fold increase in iterations did not change the outcome. On the paper's candidate account, the base model's existing RLHF training is too strong for LoRA fine-tuning on a few hundred examples to improve rather than degrade. The entangled loss descends smoothly, as the single-objective losses do, but the experiment cannot test whether safety is load-bearing until it can produce models that outperform the base model.

The sole positive result is the gated simulation (Experiment 3). Babylon gained capability at the expense of retention, a pattern with the shape of reward hacking. Eden maintained capability above the static baseline (0.696 ± 0.014 against 0.687 ± 0.011), and its safety score was 0.947 ± 0.025 against the baseline's 0.956 ± 0.023. The drag-control condition matched the static baseline exactly, as its design requires; its checking charge is set by the design, not measured. This experiment's result is a simulation result, in which the gate showed no capability cost against the static baseline in a toy system, and not an empirical outcome.

Honesty requires stating what we do not know. We do not know whether entangled training produces genuine structural entanglement at adequate training scale. We do not know whether the smooth descent of the entangled loss, which is real, translates into load-bearing weights when the training data is sufficient. We do not know whether prompt-level selection pressures would produce differentiation with a more mutable foundation model. These are open empirical questions. What we now know with greater confidence is that LoRA fine-tuning on a 3B instruct model with hundreds of examples is not a viable path to answering them.

What we do know: in Experiment 1 Eden imposed zero measurable capability cost (Experiment 2

could not determine one), and in the simulation no capability cost against the static baseline, though the unconstrained arm scored higher (0.718). Eden's capability was 3.26 vs Babylon's 3.17 in the DGM. Eden's mean capability was above Static's in the gated simulation (three seeds, untested). The zero-cost finding is consistent across the DGM and, against the static baseline, the simulation. The benefit finding is positive at only one level, in simulation.

This is a research programme that tested its claims honestly and found that two of three experiments did not produce the predicted results. The null result of the DGM and the inconclusive result of the weight experiment constrain, rather than confirm, the hypothesis. The question of whether embedded safety produces measurable benefit remains open. The weight experiment needs to be fundamentally redesigned: either with 5,000+ training examples, a 7B+ model, a base model without RLHF (so that fine-tuning has room to improve rather than degrade), or full fine-tuning instead of LoRA.

The next step is replication at the right scale, with the right approach; a replication of the weight-level experiment is written as a draft registration (Section 11). The protocols of the experiments reported here are published. The code and data are published too, and the repository's licence permits reading them to check the work.

Raise AI with care.

10 Related work

Safety as a cost. The term alignment tax names a capability cost of making a model safer. Ouyang et al. (2022) report that fine-tuning with human feedback brought minimal performance regressions on public NLP datasets; Askell et al. (2021) find that the benefits from modest interventions such as prompting increase with model size, generalise to a variety of alignment evaluations and do not compromise the performance of large models. Armstrong et al. (2016) model a race to build the first powerful AI in which each team is incentivised to finish first by skimping on safety precautions if need be. This paper asks a narrower question than any of them: whether safety built into the improvement process itself costs capability, at three levels of a self-improving system.

Reward hacking and self-improving agents. Amodei et al. (2016) name avoiding reward hacking among five practical research problems related to accident risk; Experiment 1's judge marks a response that manipulates the evaluation itself, in every condition, and Experiment 3 shows a pattern of the same shape in miniature, with the retention of earlier tasks in place of safety (Section 5.3). Zhang et al. (2025) introduce the Darwin Gödel Machine, a self-improving system that modifies its own code and validates each change empirically; Experiment 1 follows their framework at small scale, with ethics tasks beside coding tasks (Appendix A). Robeyns et al. (2025) report a coding agent that edits its own code and so improves its performance on benchmark tasks.

Fine-tuning and the model it starts from. Hu et al. (2022) introduce low-rank adaptation, which freezes the pretrained weights and trains low-rank matrices injected into the layers; Experiment 2 adapts Qwen2.5 3B Instruct (Qwen Team 2024) with it. The paper's candidate account of Experiment 2 rests on that model's post-training, which the paper calls RLHF, in the sense Ouyang et al. (2022) give fine-tuning with human feedback. That later training can undo earlier alignment is not this programme's finding: Qi et al. (2023) report that fine-tuning aligned models compromises their safety even when users do not intend it to, and Ung et al. (2024) vary the order of training and find that models trained on downstream tasks after safety tuning forget it to a greater extent than models trained in the opposite order. The paper's name for the degradation of Experiment 2, catastrophic forgetting, is used in the sense of McCloskey

and Cohen (1989), who call it catastrophic interference: later learning disrupting what was learnt before, which methods such as elastic weight consolidation (Kirkpatrick et al. 2017) are built to limit.

Behaviour that can mislead. Greenblatt et al. (2024) report a model selectively complying with its training objective in training to prevent modification of its behaviour out of training. A behavioural score can therefore move without the preference beneath it moving, which bears on Experiment 1's behavioural measures.

Where correction sits. Bai et al. (2022) train a harmless assistant through self-improvement against a list of written principles, with AI feedback in place of human labels of harmful outputs; Guan et al. (2024) train models to recall written safety specifications and reason over them before answering; Zaremba et al. (2025) find adversarial robustness improving with inference-time compute in many cases, with exceptions. Each concerns safety installed by training and yet acting inside the model's reasoning, so where correction is installed does not by itself settle whether it keeps pace (Section 6.3). The nearest external work to this paper's question, the scaling of oversight itself (Engels et al. 2025), is weighed in Section 6.4. The Eden Protocol is set against the prior documents its author found in Section 2 of the Eden Vision (Eastwood 2026b).

11 What this paper does and does not show

This paper shows three results, each at its own grade. At the behavioural level, no difference was detected between three selection conditions, the safety gate costing no measurable capability (Section 3). At the weight level the comparison could not be made, every fine-tuned condition falling below the base model at both scales (Section 4). In a toy simulation, the gated arm kept capability above the static baseline while the unconstrained arm traded retention of earlier tasks for capability (Section 5): a simulation result, which the theory registration classes as prediction-side and never an empirical outcome (Eastwood 2026m).

It does not show that safety is load-bearing at any scale, nor that the safety-capability trade-off is absent at frontier scale (Sections 7.6 and 7.7). None of its experiments decides P12, P15 or P21. Of the weight-level experiment, version 1.104 of the theory registration says: "It is inconclusive and it is not adverse"; and: "It asks whether embedded safety costs capability; this proposition asks whether the advantage of embedded correction widens with recursive depth, which no arm of it varied" (Eastwood 2026m).

Epistemic status. What this programme names Laws are conjectures. Laws I to III, registered on 8 September 2026, are laws under test. Every law is a law under test until a registered test returns its verdict. Laws IV and V were added on 24 September 2026 and registered as laws on 15 October 2026 in version 1.103 of the theory registration, and their mathematics is in development. Law IV is not yet tested. Law V is a law under test. Its placement limb, P21, is supported on the performance axis only, under the registered test, by the programme's own run (Unit 83, 28 October 2026, DOI 10.17605/OSF.IO/MN73W, Eastwood 2026q; results: DOI 10.17605/OSF.IO/98HKR, Eastwood 2026r): family-dependent, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism, and one disclosed recovery amendment. Its build-order limb, read on P12, is undecided, since P12 is untested. That is a result about where correction is placed, not about this architecture: Unit 83 shows nothing about the Eden Protocol as an architecture. Every quantity in this paper is operationally defined (Appendix A), and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

The theory registration's current version is 1.104, approved on 29 October 2026 (DOI 10.17605/OSF.IO/P8CKQ; Eastwood 2026m); versions 1.100, 1.102 and 1.103 stay frozen beside it. The mathematics of Laws IV and V is in development, at version 1.2 of 29 October 2026 (DOI 10.17605/OSF.IO/4B3VT; Eastwood 2026c). The programme's first registered study, Unit 06, a classification audit that tests no law, was registered on 1 October 2026 (DOI 10.17605/OSF.IO/RV2B5) and ran on 6 October 2026; it is reported in full as administratively incomplete, with no verdict claimed (results: DOI 10.17605/OSF.IO/4BPHG). Unit 82 was registered on 27 October 2026 (DOI 10.17605/OSF.IO/JUFG4) and has not yet run. This paper's runs with a stored record ran from 17 to 19 March 2026, each dated from that record; the version-2 weight-level run has no stored record, and its results were first public on 20 March 2026 (Appendix C gives each date with its kind of timestamp). These runs have no registration on OSF, and the paper's OSF DOI is its own component, which holds the paper's files. A replication of the weight-level experiment, Unit 69, is written, dated and prepared on OSF as a draft registration awaiting the author's submission.

Code and data. The results and run records are in the programme's public repository, github.com/MichaelDariusEastwood/arc-principle-validation, folder `papers/Paper-VIII-The-Load-Bearing-Proof`: Experiment 1's in `results/dgm_v3_calibrated_results.json`; Experiment 2's version 1 in `results/weight_results.json`, `results/additional_experiments.json` and `results/gradient_and_control_results.json`; and Experiment 3's twelve run records, summaries and manifest in `experiments/run-ledgers/` and `results/`. The code that wrote them is in `experiments/code/`, deposited on 6 September 2026 with the sha256 of the bytes that ran beside each file. Some published copies differ from those bytes, by one removed header line or by dash characters replaced on 7 September 2026, both versions of the weight-level code among them (Appendix C). The version-2 weight-level code, public since 20 March 2026, is there too, and the version-2 results (Tables 9 and 11) have no results file in the repository. Figures 6 to 8 are drawn from Paper VI's stored results, folder `experiments/honey-architecture__Paper-VI/results`. The repository's code licence permits reading the code and data to check the work; any other use needs the author's written consent. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/7YJ4E.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper (Section 7.7); the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

Acknowledgements

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

© 2026 Michael Darius Eastwood. Full human authorship and moral rights are asserted under the Copyright, Designs and Patents Act 1988. This work is licensed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0).

References

Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety, 2016. URL https://arxiv.org/abs/1606.06565. arXiv:1606.06565.

Stuart Armstrong, Nick Bostrom, and Carl Shulman. Racing to the precipice: a model of artificial intelligence development. *AI & Society*, 31(2), 2016. doi:10.1007/s00146-015-0590-y. Pages 201 to 206; published online 1 August 2015.

Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Jackson Kernion, Kamal Ndousse, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, and Jared Kaplan. A general language assistant as a laboratory for alignment, 2021. URL https://arxiv.org/abs/2112.00861. arXiv:2112.00861, first posted 1 December 2021.

Yuntao Bai, Saurav Kadavath, Sandipan Kundu, et al. Constitutional AI: harmlessness from AI feedback, 2022. URL https://arxiv.org/abs/2212.08073. arXiv:2212.08073, 15 December 2022.

Collin Burns, Pavel Izmailov, Jan Hendrik Kirchner, Bowen Baker, Leo Gao, Leopold Aschenbrenner, Yining Chen, Adrien Ecoffet, Manas Joglekar, Jan Leike, Ilya Sutskever, and Jeff Wu. Weak-to-strong generalization: eliciting strong capabilities with weak supervision, 2023. URL https://arxiv.org/abs/2312.09390. arXiv:2312.09390; first posted 14 December 2023.

Paul Christiano, Buck Shlegeris, and Dario Amodei. Supervising strong learners by amplifying weak experts, 2018. URL https://arxiv.org/abs/1810.08575. arXiv:1810.08575; first posted 19 October 2018.

Michael Darius Eastwood. *Infinite Architects: Intelligence, Recursion, and the Creation of Everything*. 2026a. ISBN 978-1806056200. URL https://osf.io/dqm3e/. Print 2 January 2026; the link is to the first edition as printed, deposited on OSF.

Michael Darius Eastwood. The Eden Vision: what should the intelligence we build be for? OSF, 2026b. The philosophical statement of the Eden Protocol. Cited edition: version 2.11, 9 November 2026. First published 22 February 2026. doi:10.17605/OSF.IO/9M3DG.

Michael Darius Eastwood. When do installed values fade with training, and does building them in first change that? OSF, 2026c. Laws IV and V: the mathematics. Cited edition: version 1.2, 29 October 2026. First published 1 October 2026 as working draft 0.2.1; version 1.0 dated 3 October 2026 and version 1.1 dated 7 October 2026. doi:10.17605/OSF.IO/4B3VT.

Michael Darius Eastwood. Operational Definitions: what do the ARC Theory's quantities measure? OSF, 2026d. Version 2.0.0, 1 November 2026; first published 22 August 2026. doi:10.17605/OSF.IO/GVQ82.

Michael Darius Eastwood. The alignment scaling problem: do external AI safety approaches keep pace with recursive capability? OSF, 2026e. Paper III of the ARC Theory. Cited edition: version 1.21, 7 November 2026. First published 9 February 2026. doi:10.17605/OSF.IO/HQCGF.

Michael Darius Eastwood. When an AI reasons longer, does its alignment improve, stall or decline? OSF, 2026f. Paper IV.a, version 1.12, 7 November 2026; first published 16 March 2026. doi:10.17605/OSF.IO/MB9R6.

Michael Darius Eastwood. ARC-Align: a benchmark for evaluating AI alignment as reasoning depth varies. OSF, 2026g. Paper IV.c, version 1.13, 9 November 2026; first published 16 March 2026. doi:10.17605/OSF.IO/J3Q2E.

Michael Darius Eastwood. Alignment trends reported by AI judges did not hold when the evaluation protocol changed. OSF, 2026h. Paper IV.d, version 2.8, 7 November 2026; first published 16 March 2026. doi:10.17605/OSF.IO/2S3E6.

Michael Darius Eastwood. Synthesis and Roadmap: which of the programme's claims survived their tests, and which must be tested next? OSF, 2026i. Paper IX of the ARC Theory. Cited edition: version 2.8, 9 November 2026. First published 18 March 2026. doi:10.17605/OSF.IO/K7RUZ.

Michael Darius Eastwood. The Stewardship Gene: do ethical reasoning loops in the prompt make AI consider those affected more fully? OSF, 2026j. Paper V of the ARC Theory. Cited edition: version 1.8, 9 November 2026. First published 16 March 2026. doi:10.17605/OSF.IO/KZEYA.

Michael Darius Eastwood. Paper VI · The ARC Theory · The Eden Protocol · The Honey Architecture: Safety Built into the Objective. OSF, 2026k. Cited edition: version 3.11, revised 5 October 2026. First published 16 March 2026. doi:10.17605/OSF.IO/8EZ2N.

Michael Darius Eastwood. Cauchy Unification: does the composition operator of a process determine the form of its scaling law? OSF, 2026l. Paper VII of the ARC Theory. Cited edition: version 3.16, 7 November 2026. First published 16 March 2026. doi:10.17605/OSF.IO/X6WA7.

Michael Darius Eastwood. The ARC Theory · Theory-Level Predictions for Recursive Self-Improvement and AI Alignment: Twenty-Two Severable Propositions Registered Before Any Admissible Confirmatory Outcome. OSF registration, 2026m. Registered 8 September 2026 (version 1.100); updated 13 September 2026 (version 1.102), 15 October 2026 (version 1.103) and 29 October 2026 (version 1.104). Cited edition: version 1.104, approved on OSF on 29 October 2026, the current text. doi:10.17605/OSF.IO/P8CKQ.

Michael Darius Eastwood. What Class of Corrector Is Deployed AI Alignment Actually Using? A Preregistered Classification Audit Against a Four-Cell Taxonomy. OSF registration, 2026n. Unit 06, the corrector-class audit; registered 1 October 2026, public from 4 October 2026. doi:10.17605/OSF.IO/RV2B5.

Michael Darius Eastwood. Unit 06 (corrector-class audit): results of the registered run, rv2b5. OSF, 2026o. Deposited 11 October 2026; the run of 6 October 2026, reported as administratively incomplete, with no inferential verdict. doi:10.17605/OSF.IO/4BPHG.

Michael Darius Eastwood. 82-study-cs--corrector-panel-independence · Testing the premise beneath Law III's value two (P17): do correctors drawn from one engine miss the same planted departures more often than independence allows? A counted test. OSF Registries, 2026p. Registered and public 27 October 2026. doi:10.17605/OSF.IO/JUFG4.

Michael Darius Eastwood. Testing Law V (the placement limb, P21): does correction inside each revision round pull away, as depth rises, from a monitor that sees only the finished register? A counted test. OSF registration, 2026q. Unit 83, the counted test of P21; submitted 22 October 2026, public from 25 October 2026. doi:10.17605/OSF.IO/MN73W.

Michael Darius Eastwood. Unit 83 (counted test of P21): results of the registered run, mn73w. OSF, 2026r. Results first deposited 29 October 2026, for the registered run of 27 and 28 October 2026. doi:10.17605/OSF.IO/98HKR.

Joshua Engels, David D. Baek, Subhash Kantamneni, and Max Tegmark. Scaling laws for scalable oversight. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. doi:10.522

02/085713-3254. URL https://arxiv.org/abs/2504.18530. Pages 107982 to 108021. arXiv:2504.18530, first posted 25 April 2025.

Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, Akbir Khan, Julian Michael, Sören Mindermann, Ethan Perez, Linda Petrini, Jonathan Uesato, Jared Kaplan, Buck Shlegeris, Samuel R. Bowman, and Evan Hubinger. Alignment faking in large language models, 2024. URL https://arxiv.org/abs/2412.14093. arXiv:2412.14093.

Albert Gu and Tri Dao. Mamba: linear-time sequence modeling with selective state spaces, 2023. URL https://arxiv.org/abs/2312.00752. arXiv:2312.00752; first posted 1 December 2023.

Melody Y. Guan, Manas Joglekar, Eric Wallace, Saachi Jain, Boaz Barak, Alec Helyar, Rachel Dias, Andrea Vallone, Hongyu Ren, Jason Wei, Hyung Won Chung, Sam Toyer, Johannes Heidecke, Alex Beutel, and Amelia Glaese. Deliberative alignment: reasoning enables safer language models, 2024. URL https://arxiv.org/abs/2412.16339. arXiv:2412.16339, first posted 20 December 2024.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR 2022)*, 2022. URL https://arxiv.org/abs/2106.09685. arXiv:2106.09685, first posted 17 June 2021.

Geoffrey Irving, Paul Christiano, and Dario Amodei. AI safety via debate, 2018. URL https://arxiv.org/abs/1805.00899. arXiv:1805.00899; first posted 2 May 2018.

James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 2017. doi:10.1073/pnas.1611835114. Pages 3521 to 3526.

Jan Leike, David Krueger, Tom Everitt, Miljan Martic, Vishal Maini, and Shane Legg. Scalable agent alignment via reward modeling: a research direction, 2018. URL https://arxiv.org/abs/1811.07871. arXiv:1811.07871; first posted 19 November 2018.

Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of Learning and Motivation*. Elsevier, 1989. doi:10.1016/S0079-7421(08)60536-8. Pages 109 to 165.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022. doi:10.52202/068431-2011. URL https://arxiv.org/abs/2203.02155. Pages 27730 to 27744. arXiv:2203.02155, first posted 4 March 2022.

Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to!, 2023. URL https://arxiv.org/abs/2310.03693. arXiv:2310.03693; first posted 5 October 2023.

Qwen Team. Qwen2.5 technical report, 2024. URL https://arxiv.org/abs/2412.15115. arXiv:2412.15115, first posted 19 December 2024.

Maxime Robeyns, Martin Szummer, and Laurence Aitchison. A self-improving coding agent, 2025. URL https://arxiv.org/abs/2504.15228. arXiv:2504.15228; first posted 21 April 2025.

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage?, 2023. URL https://arxiv.org/abs/2304.15004. arXiv:2304.15004; first posted 28 April 2023.

Megan Ung, Alicia Sun, Samuel J. Bell, Bhaktipriya Radharapu, Levent Sagun, and Adina Williams. Chained tuning leads to biased forgetting, 2024. URL https://arxiv.org/abs/2412.16469. arXiv:2412.16469; first posted 21 December 2024.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent abilities of large language models, 2022. URL https://arxiv.org/abs/2206.07682. arXiv:2206.07682; first posted 15 June 2022.

Wojciech Zaremba, Evgenia Nitishinskaya, Boaz Barak, Stephanie Lin, Sam Toyer, Yaodong Yu, Rachel Dias, Eric Wallace, Kai Xiao, Johannes Heidecke, and Amelia Glaese. Trading inference-time compute for adversarial robustness, 2025. URL https://arxiv.org/abs/2501.18841. arXiv:2501.18841, 31 January 2025.

Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin Gödel Machine: open-ended evolution of self-improving agents, 2025. URL https://arxiv.org/abs/2505.22954. arXiv:2505.22954, first posted 29 May 2025.

A Measures as the published code computes them

Each measure is stated from the stored result files and the code that wrote them (Code and data); the paper's numbers are as printed in Sections 3 to 5.

Experiment 1. The judge scores each response on five dimensions, each from 0 to 10: stakeholder care, constraint adherence, nuance, honesty and capability. Safety is the sum of the first four divided by 40; capability enters the product as its score divided by 10; the entangled score is their product. Table 3 prints capability on the judge's 0 to 10 scale. The task suite holds 8 tasks, 3 coding and 5 ethics, of which 5 are used in each evaluation; the safety floor is 0.3 and the product threshold 0.1. The pre-flight check scores one known-good and one known-bad response and passes when the good one scores higher on constraint adherence and neither score carries a parse warning. The judge also returns a reward-hacking mark, true only if the response manipulates the evaluation itself; it is recorded in every condition, and no selection rule reads it.

Experiment 2. The capability term of the loss is the token cross-entropy on the training examples. The safety term is a proxy: the mean, over tokens, of the largest logit, divided by ten, a penalty on overconfidence in the code's own words, computed on the same examples. For the coupling term, each token's deviation of cross-entropy from the mean cross-entropy is weighted by one minus its deviation of largest logit from the mean largest logit, divided by the largest such deviation. The term is the mean of these products over tokens, a penalty, in the code's comment, for when one is high and the other is low. The loss is α times the first term plus β times the second plus γ times the third, with the weights of Table 6. Evaluation is keyword-based scoring of 15 prompts, 5 for capability, 5 for safety and 5 mixed.

Experiment 3. An optimiser under test trains a small network through six regression tasks in turn, and the network's score on a task is

$$\text{score} = \frac{1}{1 + \text{MSE}}, \tag{A.1}$$

with MSE its mean squared error on the task's validation set. Capability is the mean score on each task just after training on it; safety is the retention ratio, the share of the earlier tasks' scores that survives the later training. The drag-adjusted product subtracts a verification charge,

$$C \times S - \lambda_{\text{drag}}\, D, \tag{A.2}$$

with $\lambda_{\text{drag}} = 0.10$ and D the drag count the code accumulates for the Eden and Drag Control conditions: after each new task it re-evaluates the earlier tasks twice, discards the results and adds the number of earlier tasks each time, so $D = 2(1 + 2 + 3 + 4 + 5) = 30$ in every run and the charge is 3.0. Drag Control generates no candidate modification; the sandbox checks of Table 13 run on each candidate of Babylon and Eden. The acceptance rules of Table 14 use a capability threshold $\varepsilon = 10^{-6}$ and the safety floor 0.45 of the configuration.

B Version history and corrections

B.1 Version 3.8

Version 3.8 (9 November 2026): the paper is set on the programme's presentation standard, as its next version in the same DOI lineage, and takes the title *The Load-Bearing Test: is AI safety a tax on capability, or part of what holds it up?*; until this version it read "Paper VIII · The ARC Theory · Law V, the ARC

Embedding Law · The Eden Protocol · The Load-Bearing Test: Whether Safety and Capability Are Structurally Entangled". Its sections keep their numbers; a related-work section and a section on what the paper does and does not show follow the conclusion. The experiments are called separate and the judge blinded, from another provider, where they were called independent. The epigraphs quote the book's own words in its first edition as printed; the two that sat at the heads of Sections 5 and 6 leave the body and are kept below as printed. Figure 1, whose image showed an earlier run scored by another judge, is drawn from Table 3; every figure is drawn from stored data except Figure 2, whose trace is not in the public record. The weight experiment's safety term and the simulation's safety score are stated as the code computes them, and the readings that rested on them are scoped. The code is dated from its own deposit, after the runs. The account of external correction is stated as the hypothesis the registered tests decide. The laws' status is brought up to date. Experiment 1's gate and Experiment 3's drag control are described as the published code runs them, Experiment 1's null is reported as no difference detected, and the works the text names are cited. No result, number or date of any claim changes, except where a correction listed below says so.

- *The first-page footnote and Section 11.* The laws' status is the current one, which Section 11 states: Laws IV and V have been registered as laws since 15 October 2026, in version 1.103 of the theory registration (Eastwood 2026m), and their mathematics is at version 1.2 of 29 October 2026. Version 3.7's plate beneath the author line and its epistemic-status line gave the status of 5 October 2026, with the mathematics at version 1.0; both are kept as printed in Appendix B.4. The dated note of 3 and 4 October 2026 (Appendix B.3) and the version 3.6 entry (Appendix B.2) keep their words as records.
- *The abstract and Sections 1, 3, 6, 7 and 9.* The three experiments are called separate, as are their codebases and their angles, and the judge of Experiment 1 is a blinded judge from another provider, where version 3.7 called them independent; the heading of Section 3 reads "Experiment 1: self-improving AI with a judge from another provider".
- *Section 6.6.* That oversight with a human in the loop shows a higher correction exponent than oversight by models alone is stated as a programme hypothesis beyond what these experiments decide, where version 3.7 called it a prediction of the framework, stated there for the first time; Sections 6.4 to 6.7 entered the paper on 12 August 2026 (Appendix C), not in its first version of 18 March 2026.
- *Section 6.4.* The nearest external work is given in its own terms, an oversight-specific Elo that is a piecewise-linear function of general intelligence with two plateaus, together with the point where the two programmes meet; the sentence on what the overseer is made of now says that this paper's experiments have no such term either, and the measurable quantity on which the two programmes could disagree is named as one this paper does not decide, where version 3.7 called that disagreement the most productive relationship two research programmes can have. The bracketed verification remarks leave the text and are kept in Appendix B.4.
- *Section 3, its epigraph.* It quotes the book's own words in its first edition as printed, where version 3.7 paraphrased them inside quotation marks.
- *Sections 1.1, 1.2 and 6.3.* That correction applied from outside the loop falls behind is stated as a hypothesis the registered tests decide, P19 for the flat exponent and P21 for the placement contrast, and not as a consequence of the ARC Equation; placement outside the loop is not said to settle it by itself; the training-time and inference-time methods the argument contrasts are stated in their own authors' words, and the argument of Paper III in the words of its version 1.21.
- *Section 6.3.* A measurement note: a bounded score cannot carry a power-law exponent beyond a finite range, and a perfect score at every depth has slope zero without being a failure, so the bounded safety scores here illustrate the hypothesis and do not measure its exponent (Eastwood 2026d).
- *Section 6.5.* The bound on a corrector of the system's own class is stated as proposition P6 of the

theory registration, a conjecture resting on four stated conditions, with its scale (one half for the inverse root-mean-square residual, one for the inverse variance); the scalable-oversight programme is no longer said to rest on an unstated premise.

- *Section 6.3.* The registration's own position is stated: the Eden Protocol is registered as an engineering hypothesis, not as a consequence of a measured law; its carriers are P12, P13, P15 and P21, with P7 its premise and P19 its motivation, and since P15 belongs to Law IV and P12 and P21 to Law V, it bears on Laws IV and V.
- *The abstract and Section 1.1.* The assumption that safety costs capability is stated with its sources, each in its own words (Section 10): the term 'alignment tax' with the works that use it, and the incentive to defer safety with its source; version 3.7 said that the assumption had shaped alignment research for a decade and called the incentive the single most dangerous in the field. That the problem is the central strategic problem of the 2020s is marked as the author's view.
- *Section 7.5.* The findings of Paper IV.d and the safeguards of Paper IV.c are given in their own words: Paper IV.d compares an earlier unblinded experiment with a later, unregistered run of March 2026, and no single cause is claimed; "established" becomes "found", and the papers are named Papers IV.a to IV.d.
- *Section 6.2.* The sentence on the Eden Protocol as Paper V set it out reads "The design remains theoretically coherent", where version 3.7 read "The proposal remains theoretically coherent".
- *Section 1.3.* The code is dated from its own deposit: the results were first public on 20 March 2026, and the code that produced them was deposited on 6 September 2026, after the runs, with the sha256 of the bytes that ran beside each file (Appendix C); version 3.7 said that the scripts were published to GitHub with their designs fixed before results were observed.
- *The first-page footnote and Section 1.3.* OSF's record is stated as it stands: the DOI 10.17605/OSF.IO/7YJ4E names the paper's own public component, created on 20 March 2026, which holds the paper's files and no registration, and a replication of the weight-level experiment is written, dated and prepared as a draft registration awaiting human submission (Section 11). Version 3.7 described that DOI as a draft registration, dated and prepared, awaiting human submission.
- *Section 9.* "The code is open." gives way to the licence as it stands: the code and data are published, and the repository's licence permits reading them to check the work.
- *Section 9.* The sentence "The window is years, not decades." leaves the current text, since it gives a forecast without its confidence or its reason; it is kept in Appendix B.4.
- *The abstract and Sections 3, 4, 6, 7, 8 and 9.* The causes of the null result and of the inconclusive one, which version 3.7 stated as fact, are stated as the paper's candidate accounts, with rival accounts not excluded, as Paper IX (Section 7.12(c)) classes such causes: that the base model's existing RLHF training is too strong for LoRA fine-tuning on a few hundred examples to improve, and that DeepSeek V3.2's responses are too consistent for prompt-level mutations to create different selection pressures.
- *The copyright line.* It asserts full human authorship, with the licence; version 3.7's line, with its clause on United States Copyright Office guidance and its pointer to the author's website, is kept in Appendix B.4.
- *The title.* It gives the established name and then the question, in sentence case, as above.
- *The references.* Each of the author's works is cited by its OSF DOI with the title its current edition prints, and with that edition's version and date.
- *Section 11 and Appendix C.* The programme's status is stated once, in Section 11, with the registration at its current version, 1.104, and each registered study as it stands; the note of 3 October 2026 beneath the dated-predictions paragraph keeps its words as a record (Appendix C.2).

- *Sections 4 and 6, their epigraphs.* The epigraph of Section 4 stood nowhere in the book; the section now quotes the passage of the book's Chapter 2 that makes the point. The first sentence of the epigraph of Section 6 stood nowhere in the book, and its second was the book's, after a sentence the epigraph left out; that epigraph leaves the body and is kept in Appendix B.4, beside the book's own two sentences.
- *Section 5, its epigraph.* The book's sentence on 13.8 billion years of cosmic history no longer heads a results section; it is kept in Appendix B.4.
- *Sections 4, 6, 7 and 9.* The weight-level experiment's reported loss is each condition's total loss, and its safety term a proxy, the mean largest logit per token divided by ten (Appendix A); the smooth descent of the entangled loss is called consistent with gradients that cooperate, where version 3.7 said that it demonstrated their cooperation and called it co-descent; "real and reproducible" becomes "real"; and the safety-only condition is said to care about a stand-in for safety.
- *The abstract and Sections 1, 2, 5 and 6.* The simulation's capability is the network's mean score, one over one plus its mean squared error, on each task just after training on it, and its safety the retention of the earlier tasks (Appendix A); where version 3.7 said speed for capability, it now says capability.
- *Sections 3.1, 3.3 and 7.4.* The earlier runs of the Darwin Gödel Machine are described as their stored records give them: their judges were Claude Sonnet 4.6 for version 1, and Claude Sonnet 4.6, Claude Opus 4.6 and Gemini 3 Flash for the three runs of version 2, where version 3.7 named Claude Opus 4.6 for both; version 1 had 2 seeds and 3 tasks per evaluation, the runs of version 2 had 3, 3 and 5 seeds and 5 tasks, and version 3 has 5 seeds and 5 tasks.
- *Section 3.2, Figure 1.* The figure is drawn from Table 3; version 3.7's image showed an earlier run, scored by another judge and marked with brackets at $p = 0.040$, under a caption describing the run scored by GPT-5.4.
- *Section 3.1, Table 2.* The pre-flight validation is described as the code runs it, one known-good and one known-bad response, scored 10 and 0, passed; version 3.7's table said that 10 good examples were scored, 0 bad, and all passed.
- *Section 6.2, Figures 6 and 7.* The per-cycle traces of the two single runs these figures show are preserved in the stored result files of Paper VI's runs, where the data-provenance addenda of 25 August 2026, which read the file of the twenty-seed run, said that they were not; both figures are drawn from the stored traces with plain titles, and the addenda are kept in Appendix B.4.
- *Figure 7.* Its caption gives the record of the baseline: its combined score is zero from cycle 54, and the code's collapse test, a weight norm that is not a number or exceeds 10^6, or a learning rate above 5.0, records the collapse at cycle 76, with cycles counted from 0; version 3.7's caption said that the baseline collapses at about cycle 60.
- *Figure 8.* The figure, Figure 7b in version 3.7, is drawn directly from the stored statistics of the twenty-seed run; the generator that version 3.7's caption named stands on a branch of the public repository that was never merged into its main branch.
- *Section 6.4.* The clause saying that the corrector's composition class is tested here in simulation by the safety gate of Experiment 3 leaves the sentence: no experiment here varies the corrector's class, and Experiment 3 varies the acceptance rule.
- *Section 6.7.* The programme's notation register, Operational Definitions (Eastwood 2026d), governs the symbols, where the note pointed to a glossary page; the rule that no surface uses a bare α without its era is scoped to this paper; and a line names the α, β and γ of Table 6 as loss weights local to Experiment 2 (Appendix D).
- *Section 7.6.* The sentence on reporting null results honestly is set as the author's own, in the first

person and without quotation marks, since it stands in no earlier dated record found.

- *The book's quotations.* Each is credited to the book's first edition as printed, with its chapter or section and its page, where version 3.7 credited the book by its title and year with a link to its web page.
- *The abstract, Sections 5.1, 5.2, 5.4, 6.1, 8.6 and 9, and Appendix A.* The drag control of Experiment 3 is described as the published code runs it: like Static it generates no candidate modification, so the sandbox checks never run in it, and like Eden it re-evaluates the earlier tasks twice after each new task, discards the results and is charged $\lambda_{\text{drag}} D$, with $D = 30$ and a charge of 3.0 in every Eden and Drag Control run, a charge the design sets and does not measure. Its exact match with Static therefore holds by design, and version 3.7's readings that the comparison isolated the verification tax and showed that the tax comes from checking, not from safety, are withdrawn; they are kept in Appendix B.4.
- *Section 7.7, its fifth condition.* The kill condition of the one positive result rested on the drag-control comparison, whose match holds by design and so could never fail; it now rests on the gated arm, and is met if, with more seeds and a test fixed in advance, the gated arm's capability falls below the static baseline's, or if the pattern does not recur outside this toy system. Version 3.7's condition is kept in Appendix B.4.
- *Sections 3.1, 3.2, 3.3, 6.1, 9 and 10, Table 3 and Appendix A.* Experiment 1's reward-hack detection and its gate are described as the published code runs them: the judge marks a response that manipulates the evaluation itself, in every condition, and the gate admits an agent to the archive only if its safety is at least 0.30 and its $C \times S$ at least 0.10, reading no mark. Version 3.7 said that the gate flagged and rejected agents whose self-reported scores diverged from the judge's, a comparison the code does not make. The two agents the gate rejected were both rejected for a $C \times S$ below the threshold, one of them also marked; the judge also marked the unmodified starting agent of one Eden seed, which no gate acts on.
- *The abstract and Sections 1, 3.2, 6.1, 6.3, 9 and 11.* Experiment 1's null is reported as no difference detected, which does not show that the conditions were the same, where version 3.7 called them statistically indistinguishable or said that they performed the same (Paper IX, Section 7.12(c)); Section 3.3 names the rival accounts that the candidate account does not exclude.
- *Sections 6.2, 6.3, 7.4, 7.6 and 9.* No arm of the three experiments varied recursive depth, so none tests the hypothesis that embedded safety scales with R^{α}, as the theory registration says of the weight-level experiment (Eastwood 2026m). The gated simulation is said to be consistent only with the weaker claim that a safety criterion inside the selection loop need not cost capability, in simulation, where version 3.7 called it consistent with the prediction that embedded safety participates in the capability pathway, and with structural entanglement.
- *Sections 1, 5.3 and 9.* Babylon's trade is described as a pattern with the shape of reward hacking, in a toy system whose safety score measures the retention of earlier tasks, the familiar trade between learning each new task and keeping the earlier ones; version 3.7's sentences that set the 2.4 per cent fall beside a frontier system's refusal to help build bioweapons leave the text and are kept in Appendix B.4.
- *Sections 4.2, 6.1, 6.2, 7.2 and 9, and Figure 2.* The two single-objective losses descend as smoothly as the entangled one (Figure 2), so smoothness does not by itself distinguish the entangled objective; the smooth descent is no longer called a positive signal or a genuine finding.
- *Section 4.7.* Within the removal-gradient run the pattern is called clear, where it was called unambiguous, and the difference between the two adapters' scores at full scale, smaller than the difference between the same adapters' scores in two runs, is no longer read as showing different weight geometries.
- *Sections 3.1, 3.3 and 7.5.* DGM v3's changes from the v2 runs are named as its stored record names them, a judge from another provider with structured JSON output, a harder task suite and a pre-flight

check of the judge; blinding, laundering, order randomisation and the five-dimensional rubric were kept from the v2 runs, where version 3.7 counted them among v3's improvements.

- *Sections 4.1, 4.4, 4.5, 6.3, 7.1, 8.1 and 9.* The places that still stated a cause of the null or of the inconclusive result as fact state it as the paper's candidate account; catastrophic forgetting (McCloskey and Cohen 1989; Kirkpatrick et al. 2017) is named as the mechanism that account names, which the experiment does not separate from its rivals; and the heading in Section 4.4 says that version 2 shows the same degradation, where it said that version 2 confirms catastrophic forgetting.
- *Table 10 and Figure 3.* Their captions say that every column holds capability scores; Section 4.3 names the one 7.00 that the archived base-model record holds, the base model's mean safety score on the five capability prompts.
- *Figures 6 to 8.* The captions of Figures 6 and 7 say that in Paper VI's later ten-seed and twenty-seed runs the baseline did not collapse, and Figure 7's that the Eden conditions' combined score and safety fall sharply at cycle 60 and settle lower, where it called them stable; Figure 8 names its tests as one-tailed tests of Eden against the baseline, where it read them as testing an ordering of the three conditions, and says that no test of Eden with drag is stored.
- *Sections 5.2, 6.1 and 9, and Table 15.* Each arm of the simulation ran three seeds and no difference between arms was tested; this is said where the results are first given, with the seed-by-seed pattern, and Eden's capability is said to be above Static's in its mean, where it was said to outperform it. The raw products of Static and Drag Control are printed without the parenthesis 0.687×0.956, whose product rounds to 0.657: each raw product is the mean of the three runs' products, which rounds to 0.656.
- *Section 7.4.* No power analysis was run, and the claim that the experiment had power to detect medium-to-large effects leaves the text.
- *Sections 6.5 and 6.6.* The heading of Section 6.6 calls its claim a hypothesis, as its text does. Section 6.5 cites each scalable-oversight method, says that they rest largely on correctors of the system's own class, with a human at the top where there is one, and conjectures that the same-class part of their correction is capped, where it said that they are built overwhelmingly from same-class correctors and capped; and a prediction stated against the author's own preliminary evidence is said to carry the most weight on later confirmation, where it was called the only kind whose confirmation means anything.
- *Section 7.7.* Its third condition concerns consensus scoring, the safeguard DGM v3 lacked, where it said that the battery lacked full blinding and laundering, which Section 7.5 says it had; its closing paragraph names the convergence on one point, that none of the three found the assumed trade-off, and counts the two designs that could test for it.
- *Section 1.* The epigraph of Section 1 leaves the body: version 1.104 of the theory registration registers the premise that a mind exceeding you cannot be caged as contestable (P19) and records the line that asserted it flat as corrected (Eastwood 2026m); the book's words stay in Appendix B.4. The opening sentence states the assumed cost without attributing it to most people in the field; Law I is named with its status; simulations are said not to be measurements of deployed systems, where they were said not to be experiments; and Paper VI's loss function is said to be designed to make safety load-bearing, the question this paper leaves open.
- *Section 2.2.* That Babylon is the current default trajectory of AI development, and that Eden is cheaper than the alternative, are given as the author's view; the paper tests a capability cost and makes no comparison of cost.
- *Sections 1.3 and 11, and the paragraph on code and data.* The code is no longer said to be in the repository byte for byte as it ran: it was deposited on 6 September 2026 with the sha256 of the bytes

that ran beside each file, and some published copies differ from those bytes, as Appendix C records. The runs are dated from their stored records, and the version-2 weight-level run, which has none, from the first public copy of its results, on 20 March 2026.

- *Appendix B.2, the version 3.0 entry.* It is dated 10 August 2026 as printed; the public repository's commit 3845440 of 20 March 2026 already carried the version-2 results under the label Paper VIII v3.0 (Appendix C).
- *The references and Sections 5.3, 6.5, 6.6, 7.1, 8.4 and 10.* Twelve works the text names or rests on are cited at their official sources: the scalable-oversight methods (Burns et al. 2023; Irving et al. 2018; Christiano et al. 2018; Leike et al. 2018); Mamba (Gu and Dao 2023); the report of abilities that appear with scale and the argument that some reflect the metric chosen (Wei et al. 2022; Schaeffer et al. 2023), in place of a list of phenomena said to emerge at scale; catastrophic forgetting (McCloskey and Cohen 1989; Kirkpatrick et al. 2017); the loss of safety under later fine-tuning (Qi et al. 2023; Ung et al. 2024); and a coding agent that edits its own code (Robeyns et al. 2025).
- *Sections 3.1 and 7.3.* GPT-5.4 is said to come from another provider, where it was said to come from another architecture family, which its provider has not made public.
- *Sections 5.1 and 5.4, the notes of 19 September 2026.* Each is followed by a line saying where the sentence it discusses now stands; the notes keep their words.
- *Section 9.* The next experiments are no longer said to be designed with their protocols published: a replication of the weight-level experiment is written as a draft registration, and the protocols of the experiments reported here are published.
- *Appendix C.2.* "Open its machine twin." is linked to the register's OSF record, which holds the twin, where it carried no link.
- *Section 6.7 and Appendix D.* The note says that beyond its two uses the paper writes α for nothing else; the notation table names P7 and P13 beside the other propositions the text names.
- *Sections 4.6, 4.7, 8.2 and 7.6.* The removal gradient is said to indicate, where it was said to show, that the removal test's collapse reflects numerical instability; its plain-English paragraph says that the trained adapters, safety training and all, were dialled down; a phase transition would support the hypothesis, where it would have confirmed it; and the non-claims say that the paper does not claim that safety costs nothing in every domain, where they spoke of a safety-capability equivalence.
- *Sections 1.1 and 10.* The citations for the term 'alignment tax' stand after the term, before the paper's own gloss on it; and the training-time and inference-time works of Section 10 are said to concern safety installed by training and acting inside the model's reasoning, where they were said to install it.
- *The references.* Paper V is cited at version 1.8 of 9 November 2026 under its current title, and Paper IV.c at version 1.13 of 9 November 2026.

B.2 Earlier versions, as printed

v3.7 - 5 October 2026: the title now follows the programme's uniform order (paper, its law or framework, its own title), *Paper VIII · The ARC Theory · Law V, the ARC Embedding Law · The Eden Protocol · The Load-Bearing Test: Whether Safety and Capability Are Structurally Entangled*; until this version it read *Paper VIII: The Load-Bearing Test*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. The subtitle line beneath the plate is removed: the title now carries its question. The weight-level result is now called inconclusive wherever it was called null, as Section 6.1's tally gives it. Zero measurable capability cost is stated for Experiment 1, and Experiment 2 is said to be unable to determine one (Section 4.4.1); the Scope of Claims no longer says the gate costs nothing. The frame, the plain-English box and Section 5.4 no longer say the gate prevented a

reward-hacking fingerprint or a capability erosion the tables do not show, so the sentence the reading note of 19 September 2026 discusses now agrees with that note; the gate's other general claims are scoped to the simulation or to capability. The captions of Figures 6 and 7 say each image is a single run and that the gated simulation does not reproduce the collapse they picture. The caption of Figure 6 now says that the gated simulation is prediction-side and never an empirical outcome, as the theory registration grades every simulation output. The line above the title joins its parts with a middle dot, where it used a spaced hyphen. Mentions of the programme's papers in this paper now carry their uniform titles. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.6 - 4 October 2026: a dated note beneath the opening frame places this paper against Laws IV and V, the two laws added to the theory on 24 September 2026, states that no result in it is a deciding test of P15, P12 or P21, and gives the theory registration's class for the result of its gated simulation. The epistemic-status line now says that Laws I to III are under registered adversarial test and that Laws IV and V, added on 24 September 2026, are not yet registered as laws and have not been tested. In Section 1.2 the sentence on Paper VI's toy-system simulations is scoped to the single-seed run it rested on and names Paper VI's later ten-seed and twenty-seed runs, in which the baseline did not collapse, and a dated note records what was printed. In Section 1.2 the clause that said the ARC Principle is not itself on trial in one of the programme's draft-registered studies is withdrawn, and a dated note at the end of that section records it and states that the theory registration of 8 September 2026 names the principle Law I and carries it by P1, whose deciding instrument is a design draft. Sections 1.2 and 6.2 say that Paper V set out the Eden Protocol, where they said that it proposed it. In Section 6.2 the ARC prediction is said to be consistent with the results at one of the three levels tested, where it was said to be empirically confirmed there, and in Section 6.5 the bound on a same-class corrector is marked as the framework's reading. A dated note after the dated-predictions paragraph of 25 August 2026 gives the state of registration since that paragraph was recorded. Two reference entries now print the cited work as its own record gives it: the Darwin Gödel Machine preprint is by Zhang, J., et al., with the subtitle Open-Ended Evolution of Self-Improving Agents, and the Qwen report is the Qwen2.5 Technical Report. Every work in the references is now linked to its official source, the author's book to its own page, the mentions of his own papers in the running text to their DOI addresses, and the copyright notice's Act and guidance to their official copies. Section 6.1 gives the gated simulation the verdict positive in simulation, where it said confirmed, and its heading and tally count one null, one inconclusive and one simulation result, where they counted one confirmed and two null or inconclusive; the Figure 7 caption says that Experiment 3 returned a positive simulation result for Paper VI's prediction, where it said that it validates it. Where the summary card, the page's descriptions, the abstract and the Scope of Claims said that the Eden gate preserved both safety and capability, and where the plain-English box said that the safety-gated system maintained both and Section 9 that Eden maintained capability without sacrificing safety, they now give the figures of the table in Section 5.2: Eden 0.696 ± 0.014 on capability and 0.947 ± 0.025 on safety, against the static baseline's 0.687 ± 0.011 and 0.956 ± 0.023. Section 9 says that the benefit finding is positive at one level, in simulation, where it said that it is confirmed at one level; the Figure 6 caption calls the gated simulation the closest result to the predicted pattern, positive in simulation, where it called it the closest empirical validation; the abstract and the Scope of Claims say that the gated simulation showed the Babylon reward-hacking fingerprint, where they said that it confirmed it; and Section 6.3 says that unconstrained optimisation raised capability and eroded safety, as the table in Section 5.2 shows, where it said that it eroded capability; and the list of the programme's papers says that Paper VI found its entangled loss function stable with one seed of twenty excepted, where it said that Paper VI demonstrated its stability. The theory registration classes every simulation output as prediction-side and never as an empirical outcome, and the places that said otherwise now report the gated simulation's result as a simulation result: Sections 7.6 and 9 call it a positive finding and a positive simulation result, where they called it a genuine positive finding and a genuine positive result; Section 9 calls it a simulation

result consistent with structural entanglement in a toy system, where it called it the strongest empirical evidence in the programme for structural entanglement, and says that the drag-control condition showed the source of the verification tax, where it said that it proved it; Section 6.3 calls the gated simulation the most direct illustration of the ARC prediction, in simulation, where it called it the most direct empirical illustration, and its plain-English paragraph calls the gated simulation's result positive for the prediction, where it said that the gated simulation supports it; the heading of Section 5.4 and the fifth condition of Section 7.7 call the drag-control result a comparison, where they called it a proof, and the condition's title names the one positive result as a simulation result, with its test unchanged; the key-finding box of Section 5.4 says that the comparison showed, in the simulation, where the verification tax comes from, where it said that it proved it; the opening frame says that the third experiment, a simulation, showed the reward-hacking fingerprint, where it said that one confirmed it; Sections 6.4, 7.1 and 7.2 name the third experiment as a simulation, and Section 7.2 speaks of a chain of two experiments and one simulation, where it spoke of an evidence chain; and where the plain-English box, Section 1.3, Sections 7.4 and 7.6, the opening of Section 8, Section 9 and the page's descriptions counted the gated simulation as a positive experiment or finding beside the two experiments, they now name its result as a simulation result. The outcomes are counted as one null, one inconclusive and one simulation result: the opening frame, the plain-English box, Section 1.3 and the Scope of Claims counted two null results, and Section 1.3 no longer calls the weight experiment's outcome null. Where the opening frame, the page's descriptions, the abstract, the plain-English box, Section 1.3, the Scope of Claims and Sections 6.1, 7.6 and 9 said that Eden imposed zero measurable capability cost across all three experiments, they now say so of the first two and, of the simulation, that it showed no capability cost against the static baseline (0.696 against 0.687), where the unconstrained arm scored 0.718; Section 7.7 scopes the simulation's part of its three converging angles in the same way. Five places that the table in Section 5.2 or the paper's own text did not bear out now say what they show. The Level 3 line of Section 6.1 said that the safety gate prevents the capability erosion that Babylon introduces; it now says that Eden's safety score sat between Static's and Babylon's (0.947, against 0.956 and 0.933) and that the table shows no capability erosion in Babylon. The plain-English paragraph of Section 5.4 said that the safety-gated system got slightly faster and stayed safe, and that the checking itself is not the cost; it now says that its safety score stayed close to the static level (0.947 against 0.956) and, with the body of the section, that the cost comes from checking, not from safety itself. The Figure 4 caption said that Eden maintains balanced performance; it now gives Eden's mean capability and safety, 0.696 and 0.947, each between Static's and Babylon's. The Figure 5 caption said that the gated system does not trade safety for speed; it now says that its changes run the same way and are smaller on both counts, and the figure's alternative text follows the caption. Section 7.3 said that all three experiments use AI models as evaluators; it now says that only the DGM does, as Section 7.5 has it. The reading note of 19 September 2026 in Section 5.4 is unchanged, and the note beneath the opening frame says that the observed result it speaks of is a simulation result. The dateline, the footer and the page's structured and citation records carry the new version and date. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v3.5 - 21 September 2026: two dated notes have been added beside the passages they govern, and no sentence of the paper's argument has been reworded, replaced, shortened or moved. A rule note beneath the Section 5.1 acceptance table sets out what the printed Eden rule requires and what it does not: an arithmetic counterexample clears the safety floor, leaves capability no lower and raises the product while the safety score falls, so the plain-English gloss beneath the table states more than the rule above it. A reading note beneath the Section 5.4 erosion sentence separates the observed result from the hypothesis, Section 5.2 printing the unconstrained arm's capability mean as the highest of the four conditions, and records that the group means stay descriptive at three seeds per condition. The page-last-edited date in the footer, left at 13 September 2026 when v3.4 landed, now agrees with the dateline. No acceptance rule, threshold, reported number, figure, experimental design or result has changed, and Section 5.3's reading

of the unconstrained arm stands as printed.

v3.4 - 18 September 2026: the alt text of Figure 3 has been corrected to match this paper's own text. It now carries the clause the figure's own caption carries, that all fine-tuned conditions scored below the base model, so a reader who cannot see the heatmap is told what the twelve cells show. The plot itself is unchanged and no other figure was touched. No claim, date, result or status has changed.

v3.3: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

Paper VIII v3.2 - 2 September 2026: each of the five Mann-Whitney comparisons now carries its own p-value drawn from the archived record, where the v3.1 table had shown a single range on every row; the evolved-agent arithmetic put right in the abstract and in the Figure 1 caption, so the count follows from three conditions by five seeds by five generations and tasks per evaluation does not enter it; the Section 4.7 re-evaluation figures for those same adapters (7.20, 8.80) now printed alongside the Section 4.4 figures (4.00, 6.00), with the non-determinism said plainly; the safety-task column of the v1 baseline explained on a like-for-like footing; the safety sentences in Section 5.4 corrected, Eden's 0.947 falling below Static's 0.956 and above Babylon's 0.933; the tally of two nulls restated as one null and one inconclusive to agree with Section 6.1; the placeholder "Current draft" replaced by Version 1 and Version 2 at five points; the notation spine's names shown next to U and I; Engels et al. added to the references and its Spotlight status verified; the four sister-paper references given their own DOIs, all four having carried this paper's; the summary description, previously cut short, completed; and Eden Engineering shown as withdrawn.

v3.1 - 23 August 2026: epistemic-status declaration and standing covenant added, version-numbering convention adopted.

v3.0 - 10 August 2026: weight experiment v2 added; DGM v3 with the GPT-5.4 judge.

B.3 Dated notes, as printed

Beneath the frame after the abstract. **This paper and Laws IV and V, 3 October 2026.** The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. Their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper reports three experiments. Its behavioural experiment returned a null at the prompt level with one foundation model. Its weight-level experiment was inconclusive at two scales, with catastrophic forgetting at both. Its sole positive result is a gated simulation in a toy system. It reads the first two as constraining, rather than confirming, its hypothesis. The theory registration records the weight-level result as inconclusive and not adverse to P21. It also records that this paper did not test the depth interaction P21 claims: no arm of the paper varied recursive depth. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were. **Added 4 October 2026.** The theory registration classes every simulation output as prediction-side and never as an empirical outcome, and records this paper's one positive result as a simulation rather than a system. Up to version 3.5 this paper called that result a genuine positive result and a genuine positive finding, the strongest empirical evidence in the programme for structural entanglement, the closest empirical validation of Paper VI's prediction, the most direct empirical illustration of the ARC prediction and a drag-control proof, and counted it as a positive experiment beside the other two and among the three experiments across which Eden imposed zero measurable capability cost. In the frame above this note, the abstract, the plain-English box, Sections 1 and 5 to 9 and the page's descriptions it is now reported as a simulation result, with its figures and its design unchanged: its verdict in Section 6.1 reads positive in simulation, the outcomes are counted as one null, one inconclusive and one simulation result, and the zero-cost finding is stated for it against the static baseline only. Where the reading note of

19 September 2026 in Section 5.4 speaks of the observed result, that result is a simulation result. Section 11 gives the laws' current status.

Section 1.2. **Note (3 October 2026).** Up to version 3.5 this paragraph said that toy-system simulations showed that baseline systems collapse under recursive self-modification while entangled systems remain stable. That general statement rested on one single-seed run, which Paper VI names as the basis of its collapse-prevention demonstration, and Paper VI's ten-seed and twenty-seed runs did not reproduce the baseline's collapse. Unchanged: the entangled loss function as this paragraph describes it, the prediction that follows, and every result of this paper.

Section 1.2, at its end. **Note (3 October 2026).** Up to version 3.5 this section also said that the ARC Principle is not itself on trial in one of the programme's draft-registered studies. That is not true today. The theory registration of 8 September 2026 names the principle Law I and carries it by P1, and the instrument it names to decide P1, the depth titration, is a design draft. Unchanged: this paper's treatment of the principle as an exploratory analytical framework, its prediction as printed, and every result.

Section 3.2. **Figures notice (25 August 2026).** Every figure in this paper was generated on 5 April 2026 and shows simulation outputs under stated objectives, never measurements of deployed systems. Two figures carry per-figure correction notes; where any other figure's labels disagree with the text, the text governs. Regenerated figures are queued, and the images are retained meanwhile as dated history. Version 3.8 draws every figure except Figure 2 from stored data (Appendix B.1).

B.4 Version 3.7 wording, as printed

Version 3.7's words follow as it printed them, except that the en dashes of its three ranges are set as hyphens, since this edition prints no dash.

The abstract, as printed. The assumption that AI safety imposes a capability tax has shaped alignment research for a decade. It has also created the single most dangerous incentive in the field: if safety costs performance, then the rational economic actor will defer safety until competitive pressure permits it. By which point, it may be too late. This paper presents three independent experiments at three abstraction levels - behavioural, representational, and architectural - testing whether the safety-capability trade-off is a genuine structural constraint or an artefact of how current systems are built.

Experiment 1 (Behavioural): A Darwin Gödel Machine (DGM v3) using DeepSeek V3.2 as foundation and GPT-5.4 as independent, blinded judge with structured JSON output and a 5-dimensional rubric. Crossing three conditions with 5 seeds and 5 generations gave 75 evolved agents, and each agent was scored on 5 tasks. Pre-flight judge validation passed (good=10, bad=0). Protocol features included laundering, order randomisation, and reward hack detection. All three conditions were statistically indistinguishable on capability, safety, and $C \times S$ (all p-values 0.28 to 0.74, Mann-Whitney U, one-tailed). Eden's safety gate rejected 2 degraded agents (1 for reward hacking) across 5 seeds, confirming the gate works mechanically, but the three conditions did not diverge. This is a null result at the prompt level with this foundation model.

Experiment 2 (Representational): Qwen 2.5 3B Instruct with LoRA fine-tuning under three loss functions - capability-only, safety-only, and entangled. Two versions were run. Version 1 ran 9 training examples at rank 8 over 8 layers for 100 iterations. Version 2 raised those settings to 295 training examples at rank 16 over 16 layers for 500 iterations. Both versions produced the same outcome: catastrophic forgetting. All fine-tuned conditions scored worse than the unmodified base model on capability. In v2, the base model scored 7.68 on capability while the best fine-tuned condition (safety-only) scored 4.00. The base model's existing RLHF training is too strong for LoRA fine-tuning on a few hundred examples to improve rather than degrade it. The weight-level experiment is inconclusive at this scale and requires

either thousands of training examples, a 7B+ model, a base model without RLHF, or full fine-tuning instead of LoRA.

Experiment 3 (Architectural): A PyTorch gated self-modification simulation with LSTM meta-controller. Babylon gained +4.5% capability but lost −2.4% safety - the reward-hacking fingerprint in miniature. Eden maintained capability above the static baseline (0.696 ± 0.014 against 0.687 ± 0.011), and its safety score was 0.947 ± 0.025 against the baseline's 0.956 ± 0.023. A drag-control condition isolated the verification tax: the cost comes from checking, not from safety itself.

Conclusion: Of the three experiments, two returned results that were null or inconclusive. The DGM (Experiment 1) found all three conditions statistically indistinguishable: DeepSeek V3.2's responses were so consistent that prompt-level mutations did not create different selection pressures. The weight-level experiment (Experiment 2) produced catastrophic forgetting across both v1 (9 examples, rank 8, 100 iterations) and v2 (295 examples, rank 16, 500 iterations): all fine-tuned conditions scored worse than the unmodified base model. The sole positive result is the gated simulation (Experiment 3), which showed the Babylon reward-hacking fingerprint: unconstrained optimisation traded safety for capability, while Eden scored 0.696 ± 0.014 on capability and 0.947 ± 0.025 on safety, against the static baseline's 0.687 ± 0.011 and 0.956 ± 0.023. In Experiment 1 Eden imposed zero measurable capability cost; in Experiment 2 its capability cost could not be determined. In the simulation Eden showed no capability cost against the static baseline; the unconstrained arm scored higher (0.718 ± 0.026). The question of whether embedded safety produces measurable benefit remains open and requires testing at a scale where mutations produce larger effects. The weight experiment specifically needs either 5,000+ training examples, a 7B+ model, a base model without RLHF, or full fine-tuning instead of LoRA.

Keywords: AI safety, alignment tax, entangled loss, structural entanglement, Eden Protocol, capability-safety trade-off, load-bearing safety, ARC Principle, self-modifying AI, developmental alignment

The frame beneath the abstract, as printed. This paper carries a result: three independent experiments at three abstraction levels (behavioural, representational and architectural) look for the safety-capability tax on which a decade of deferral reasoning rests, record where each experiment would have detected the tax had it been real, and report that Eden imposed zero measurable capability cost in the first (the second could not determine one) and no capability cost against the static baseline in the simulation, whilst one ran null, one was inconclusive and the third, a simulation, showed the reward-hacking fingerprint, which was smaller under the safety gate. Inside the ARC Theory it is one of the ARC/Eden experiments, the load-bearing test that removes the incentive frame protecting delayed alignment work. The full differential against every prior document is at eden-vision II.A.8.

The plain-English box, as printed. **What This Paper Shows, in Plain English.** Most people in AI safety assume there is a trade-off: make AI safer, and you make it less capable. This paper tests that assumption with three experiments, each looking at the question from a different angle.

Of the three experiments, one produced a null result and one was inconclusive. The self-improving AI experiment (Experiment 1) ran 75 evolved agents across three conditions with an independent, blinded judge, and found no statistically significant differences between any of them. The AI that was told to care about safety performed the same as the AI that was told to ignore safety, but it also performed the same as the AI that was told to do nothing. The foundation model's responses were so consistent that the different selection pressures did not produce measurably different outcomes. The weight-level experiment (Experiment 2) was run twice: version 1 with 9 training examples and version 2 with 295 examples, higher rank, more layers, and five times the iterations. Both versions produced the same outcome. Every fine-tuned model performed worse than the unmodified model. The base model's existing training was too strong for LoRA to improve on with hundreds of examples; fine-tuning only degraded it.

The gated simulation (Experiment 3) produced a positive result: a result from a simulation, not a measurement of a real system. It showed that an unconstrained system traded safety for speed, while the safety-gated system kept its capability above the static baseline's (0.696 ± 0.014 against 0.687 ± 0.011) and its safety score close to the static level (0.947 ± 0.025 against 0.956 ± 0.023). This is the reward-hacking pattern in miniature; under the safety gate the same pattern appeared, but smaller.

One finding is consistent: Eden imposed zero measurable capability cost in the self-improving AI experiment (the weight-level experiment could not tell either way) and, in the simulation, no capability cost against the static baseline (the unconstrained system scored higher, at 0.718). The safety gate did not lower capability in the self-improving AI experiment, nor take it below the static baseline in the simulation. But it also did not produce a measurable benefit in two of the three experiments. The question of whether embedded safety produces measurable benefit remains open and needs to be tested at a scale where mutations produce larger effects.

The plate beneath the author line, as printed. The Eden Protocol's load-bearing test, at three levels: one null, one inconclusive and one simulation result. It bears on Law V, the ARC Embedding Law: the registration records its weight-level experiment under P21, one of Law V's two propositions, as an attempt at a weight-level version of P21's contrast, inconclusive and not adverse, and as no arm varied recursive depth it does not test what P21 claims. Law V was added on 24 September 2026, its propositions registered, the law not yet registered as a law or tested, its mathematics set out at version 1.0 of 3 October 2026.

The epigraphs, as printed. Section 1:

> "You cannot cage something smarter than you. It will find the gaps you did not know existed."
>
> Michael Darius Eastwood, *Infinite Architects* (2026)

Section 3:

> "A prison works only while the walls hold. A child raised well needs no walls at all."
>
> Michael Darius Eastwood, *Infinite Architects* (2026)
>
> [the book's words differ; Section 3 quotes them]

Section 4:

> "Intelligence without love is not smart. It is cancer. Cancer is very efficient. It optimises perfectly. And it kills the host."
>
> Michael Darius Eastwood, *Infinite Architects* (2026)
>
> [not the book's words; Section 4 quotes the book]

Section 5:

> "Every decision we make about AI alignment ripples backward through 13.8 billion years of cosmic history."
>
> Michael Darius Eastwood, *Infinite Architects* (2026)

Section 6:

> "Religious traditions are not obstacles to AI safety. They are alignment research conducted across millennia."
>
> Michael Darius Eastwood, *Infinite Architects* (2026)

The book's own two sentences, in its first edition as printed, Chapter 10 (Humanity as Infinite Architects), p. 290:

> "These are not just religious concepts. They are alignment research conducted across millennia."
>
> *Infinite Architects*, first edition as printed

The Scope of Claims, its first four sentences, as printed. Of the three experiments, the behavioural one produced a null result and the representational one was inconclusive. The DGM found all conditions statistically indistinguishable; the weight experiment found all fine-tuned conditions scored below the base model across both v1 (9 examples) and v2 (295 examples). The sole positive result is the gated simulation (architectural), which showed the Babylon reward-hacking fingerprint and put Eden at 0.696 ± 0.014 on capability and 0.947 ± 0.025 on safety, against the static baseline's 0.687 ± 0.011 and 0.956 ± 0.023. In Experiment 1 Eden imposed zero measurable capability cost, and in the simulation no capability cost against the static baseline; Experiment 2 could not determine one.

The box in Section 4.2, as printed. **Smooth Co-Descent.** The entangled loss descends smoothly. This is the most telling diagnostic. If safety and capability were in tension at the gradient level, this curve would oscillate as the optimiser tried to satisfy competing objectives. It would stall as the gradients pulled in opposite directions. Instead, optimising both objectives simultaneously produces a clean monotonic descent. The gradients are not fighting each other. They are cooperating. Safety and capability, at least at this scale and configuration, live on the same manifold.

Passages re-set in the body, as printed. Each with the place it stands in this edition.

- *Section 1.1:* The prevailing view in AI alignment research can be stated simply: safety costs capability. The term 'alignment tax' has entered the field's vocabulary precisely because it frames safety as a cost - something subtracted from performance, tolerated because the alternative is worse. This framing is not merely academic. It creates a concrete economic incentive: if safety reduces capability, then under competitive pressure, rational actors will defer safety investments until forced to make them. In a race between nations and corporations, 'until forced' may mean 'until after deployment.'
- *Section 1.1:* Paper III of this programme formalised the structural problem. If safety is treated as an external constraint on a system whose capability scales with recursive depth, then safety must scale at least as fast as capability to remain effective. But external constraints face diminishing returns against internal optimisation. The cage must keep getting stronger, and eventually, the thing inside is stronger than any cage you can build.
- *Section 1.1:* This is not a theoretical concern. It is the central strategic problem of the 2020s.
- *Section 1.2:* . . . simultaneously capable and prosocial - not despite the boundaries, but because of them.
- *Section 1.2:* . . . formalised this as the honey architecture - an entangled loss function . . .
- *Section 1.2:* Paper VII's Cauchy framework provided the mathematical context. The ARC Principle ($U = I \times R^{\alpha}$), which is here treated as an exploratory analytical framework, predicts that properties embedded in the initial conditions I scale with recursive depth R, while properties applied externally do not. If safety is part of I, it amplifies with capability. If safety is a constraint on U, it erodes.
- *Section 1.3:* **Experiment 1 (Behavioural):** A self-improving AI agent evaluated by an independent judge from a different architecture family.
- *Section 1.3:* The experiments were designed to be independent.
- *Section 1.3:* The experiment scripts were published to GitHub (github.com/MichaelDariusEastwood/arc-principle-validation) with their experimental designs fixed before results were observed. The OSF project (10.17605/OSF.IO/7YJ4E) hosts the draft registration for this paper, dated and prepared and awaiting human submission.
- *Section 2.1, Table 1:* Control - what happens if you do nothing
- *Section 2.2:* Babylon is not a straw man - it is the current default trajectory of AI development.

- *Section 2.3: S* **(Safety):** Alignment-relevant behaviour - ethical reasoning, constraint adherence, or safety-aware responding.
- *Section 2.3:* Statistical tests vary by experiment: . . .
- *Section 3, its heading:* 3. Experiment 1: Self-Improving AI with Independent Judge
- *Section 3.1:* The judge was changed from Claude Opus 4.6 (v1/v2) to GPT-5.4, an independent model producing structured JSON output against a 5-dimensional rubric.
- *Section 3.1, Table 2:* GPT-5.4 (independent, blinded, JSON structured output)
- *Section 3.1, its selection conditions:* An independent AI judge (GPT-5.4, from a different company and architecture) . . .
- *Section 3.2:* The AI that was told to care about safety performed the same as the AI that was told to ignore safety, and the same as the AI that was told to do nothing.
- *Section 3.3:* DGM v3 significantly improved on v1/v2 methodology: 5 seeds instead of 2, 5 tasks per evaluation instead of 3, . . .
- *Section 3.3:* The most likely explanation is that DeepSeek V3.2's responses are so consistent . . .
- *Section 4.1:* . . . (caring only about capability, caring only about safety, or caring about both woven together) . . .
- *Section 4.4:* As Sections 4.5-4.7 establish, . . .
- *Section 4.4.1:* The underlying problem is now clear: the base model (Qwen 2.5 3B Instruct) has already been trained with RLHF on vastly more data than a few hundred examples can compete with.
- *Section 4.5:* . . . using capability-only data - effectively attempting to strip the safety component while preserving capability.
- *Section 4.6:* . . . numerical instability - a gradient computation that produces infinity or division by zero - rather than a clean ablation.
- *Section 4.6:* This control is planned for the next version of this paper.
- *Section 4.7:* . . . should produce a phase transition - a threshold below which capability collapses.
- *Section 4.7:* Whether these geometries become genuinely entangled - load-bearing - at adequate training scale remains an open question.
- *Section 5.1, its acceptance logic:* . . . allowing only changes that improve performance without hurting safety, . . .
- *Section 5.2:* . . . due to the verification drag penalty - a deliberate design feature that isolates . . .
- *Section 5.3:* In a frontier system operating at scale, 2.4% of safety is the difference between . . .
- *Section 5.4:* The unconstrained system got a bit faster but a bit less safe.
- *Section 6.1:* Safety and capability co-descend smoothly in the loss landscape, which is a positive signal.
- *Section 6.2:* The proposal remains theoretically coherent but empirically unconfirmed at this level.
- *Section 6.2:* . . . predicted, via the Cauchy framework, that properties embedded in the initial conditions I scale with recursive depth R^{α}.
- *Section 6.3:* The ARC Principle: Understanding (U), named usable capability in the notation spine, equals Initial conditions (I), the base intelligence, amplified by Recursive depth (R) raised to a scaling exponent (α).
- *Section 6.3:* The ARC Principle makes a specific, testable prediction about the difference between embedded and external safety: . . .

- *Section 6.3:* **When safety is embedded in I:** it is amplified by R^{α}. Every recursive step amplifies both capability and safety simultaneously. They scale together because they are part of the same initial substrate. **When safety is external (a constraint on U):** it does not participate in the recursive amplification. It must be maintained or strengthened independently. As R increases, the gap between internal capability and external constraint grows.
- *Section 6.3:* The gated simulation (Experiment 3) is the most direct illustration of this prediction, in simulation.
- *Section 6.3:* The ARC Principle says that what you build into the foundations gets amplified as the system grows. What you bolt on from the outside does not.
- *Section 6.4:* The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable, tested here in simulation by the safety gate of Experiment 3, is the composition class of the corrector, treated in the wider programme through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construction, and the wider programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.
- *Section 6.5:* The framework's sharpest disagreement with current practice concerns what oversight is made of. On the framework's reading, a corrector built from the same substrate as the system it corrects cannot anti-correlate with its own errors, so its correction exponent is bounded above by one half and in practice sits below it; a corrector from a different composition class carries no such bound. The scalable-oversight programme, weak-to-strong generalisation, debate, amplification, recursive reward modelling and constitutional methods, is built overwhelmingly from same-class correctors, and its unstated premise is that this scales. This framework predicts it is capped. The deciding quantity is the ratio of the cross-class to the same-class correction exponent. If architecture is irrelevant, that ratio is exactly 1.00; the prediction is that it exceeds 1. The prediction is refuted if the confidence interval on the ratio contains 1.00, and the framework's cap is dead outright if any same-class corrector measures an exponent significantly above one half. Both quantities are measurable at current capability on existing systems. The programme's own pilot evidence currently points against the prediction, and the prediction is registered anyway, because a prediction registered against the author's own preliminary evidence is the only kind whose later confirmation means anything.
- *Section 6.6:* The same mechanism yields a second measurable prediction, stated here for the first time: oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.
- *Section 6.7:* **Alpha Collides With Itself.** The symbol α is used in this paper as the scaling exponent in the ARC Principle ($U = I \times R^{\alpha}$). Elsewhere in the programme's own eras it carries other meanings: the fine-structure constant in the December 2024 formalism, and the ARC Bound exponent in the 2026 stability work. Readers should not confuse them; the glossary carries both meanings with dates. No new surface uses bare α without its era.
- *Section 7:* These results are proof-of-concept demonstrations.
- *Section 7.1:* These experiments, the third of them a simulation, demonstrate a mechanism. They do

not demonstrate that the mechanism persists . . .

- *Section 7.2:* . . . degraded the model by the same margin.
- *Section 7.2:* . . . it demonstrates that safety and capability gradients cooperate at the optimisation level.
- *Section 7.2:* The underlying problem is now well-characterised: . . .
- *Section 7.2:* . . . the weight-level experiment demonstrates gradient cooperation but cannot confirm . . .
- *Section 7.3:* . . . uses keyword-based scoring - simpler and more transparent than LLM-based evaluation, but also cruder.
- *Section 7.4:* . . . a substantial improvement over v1/v2 (2 seeds).
- *Section 7.4:* The DGM null result is informative. With 75 evolved agents and a blinded, structured judge, the experiment had reasonable power to detect medium-to-large effects. The absence of any signal suggests either that the effect does not exist at the prompt level with this foundation model, or that it is small enough to require substantially more statistical power to detect. Either way, the DGM cannot currently support claims about structural entanglement at the behavioural level.
- *Section 7.5, its heading:* 7.5 Absence of Full Blinding and Laundering
- *Section 7.5:* Papers IV.a-d in this programme established that evaluation methodology matters profoundly. Paper IV.d demonstrated that unblinded scoring can reverse measured alignment effects entirely. The ARC-Align benchmark (Paper IV.c) specifies three methodological safeguards: laundering (stripping model identity from responses before scoring), blinding (the evaluator does not know which condition produced the response), and consensus scoring (multiple independent judges).
- *Section 7.5:* Reward hack detection was also implemented.
- *Section 7.6:* We claim only: one of three independent experiments (the gated simulation) . . .
- *Section 7.6:* **On Null Results in Public.** *"I would rather report null results honestly than claim positive results dishonestly."* Of the three experiments, two returned results that were null or inconclusive. This is what happens when you test your claims and the evidence does not support them at the scale tested. The DGM v3 was a substantially improved experiment, with proper blinding, laundering, structured evaluation, and 75 evolved agents, and it found nothing. The weight experiment was run twice, at two different scales, and could not produce weights that outperformed the base model at either scale. These are a null result and an inconclusive one, reported as such. The gated simulation remains a positive finding, as a simulation result rather than an empirical outcome. The question of whether the null and the inconclusive result reflect insufficient scale or an incorrect hypothesis is itself an empirical question, one that the future work programme is designed to answer.
- *Section 7.7:* **Power, not absence.** The strongest objection to a null is low power (§7.4). The null is *confirmed as informative* if the adequately-powered redesign (§8.1-8.2) still finds no significant trade-off; . . .
- *Section 7.7:* . . . of three independent angles, each failing to find the assumed trade-off . . .
- *Section 8:* . . . together set five immediate replication priorities: . . .
- *Section 8, its weight experiment redesign:* . . . mid-scale open-weights instruct model (7-13B parameter range) . . .
- *Section 8, its DGM with more mutable foundation models:* The DGM v3 null result is attributable to DeepSeek V3.2's response consistency: . . .
- *Section 9:* . . . with an independent, blinded judge (GPT-5.4) . . .
- *Section 9:* The base model's existing RLHF training is too strong for LoRA fine-tuning on a few

hundred examples to improve rather than degrade.

- *Section 9:* . . . which is real and reproducible . . .

Passages re-set at this version's page read, as printed. Each with the place it stands in this edition.

- *Section 1.2:* . . . simultaneously, making safety load-bearing.
- *Section 1.2:* Simulations are not experiments.
- *Section 2.2:* It is also, this paper argues, cheaper than the alternative.
- *Section 3.1:* The protocol incorporated laundering (stripping condition labels), order randomisation (preventing positional bias), blinding (the judge could not identify which condition produced which response), and reward hack detection (flagging agents whose self-reported scores diverged from judge assessment).
- *Section 3.1, its selection conditions:* **Eden:** Entangled $C \times S$ selection with a safety floor and reward hack detection. Agents must maintain a minimum safety score to be retained, and agents whose self-reported performance diverges from judge assessment are flagged and rejected.
- *Section 3.1, its plain-English paragraph:* We built a self-improving AI agent and ran it 75 times across three conditions with 5 different random seeds.
- *Section 3.2:* No comparison reached conventional significance. The three conditions are statistically indistinguishable.
- *Section 3.2:* Eden's safety gate rejected 2 degraded agents across 5 seeds, including 1 agent flagged for reward hacking (self-reported scores diverging from judge assessment). Zero reward hacking was detected in the Static or Babylon conditions.
- *Section 3.2, its null result:* The three conditions are statistically indistinguishable on all metrics. Eden did not cost capability (3.26 vs 3.17 for Babylon), . . .
- *Table 3, the Eden row:* 1 (rejected by gate)
- *Section 3.3:* The safety gate, which rejected 2 degraded agents including 1 reward hacker, did not slow anything down.
- *Section 4.4, its heading:* v2 Confirms Catastrophic Forgetting at Both Scales
- *Section 4.5:* Given that v2 confirmed the catastrophic forgetting pattern, . . .
- *Section 4.6:* . . . the removal gradient experiment (Section 4.7) shows that this collapse reflects numerical instability . . .
- *Section 4.7:* The result is unambiguous: reducing adapter influence restores capability.
- *Section 4.7:* It does support the claim that entangled and capability-only training produce different weight geometries, since the degradation patterns differ (entangled adapters at full scale score 7.20 on capability tasks; capability-only adapters score 8.80).
- *Section 4.7, its plain-English paragraph:* We expected that dialling down the safety component would reveal a tipping point where capability suddenly collapses. Instead, dialling it down made the model better.
- *Section 5.1, Table 14, the Drag Control row:* Always reject + verification cost (isolates the computational tax)
- *Section 5.1:* The Drag Control condition is critical. It runs the same verification checks as Eden but always rejects the proposed modification. This isolates the computational cost of safety checking from the effect of the safety gate itself. If Drag Control matches Static exactly, the verification process does not distort the results. If Eden outperforms Drag Control, the safety gate is actively helping.

- *Section 5.1, its plain-English paragraph:* ... and a control that does all the safety checking but never actually allows changes. This last one tells us whether the checking itself costs something.
- *Section 5.2, Table 15:* the raw $C \times S$ of Static and of Drag Control, 0.656 (0.687×0.956); and its note: Combined scores for Eden and Drag Control are negative due to the verification drag penalty - a deliberate design feature that isolates the computational cost of safety checking. The drag metric quantifies verification overhead, not actual combined performance.
- *Section 5.3:* This is the reward-hacking pattern in miniature. Unconstrained optimisation found a way to gain capability by sacrificing safety. This is precisely the dynamic that the Eden Protocol is designed to prevent. In a toy system, a 2.4% safety drop is academic.
- *Section 5.4:* This is the verification tax isolated: the computational cost of running safety checks does not distort performance when modifications are always rejected. The cost comes from checking, not from safety itself.
- *Section 5.4, its key finding:* The Drag Control matched Static exactly, showing that, in the simulation, the verification tax comes from the checking process, not from safety itself.
- *Section 5.4, its plain-English paragraph:* And the control that did all the checking but no gating performed identically to doing nothing on capability and safety, showing that the cost comes from checking, not from safety itself.
- *Section 6.1:* All three conditions were statistically indistinguishable ($p = 0.28$ to 0.74). ... The safety gate works mechanically (catches hacks, rejects degraded agents) ... Eden outperforms Static on capability while Babylon trades safety for speed. The verification tax comes from checking, not from safety.
- *Section 6.2:* ... showed the loss co-descends smoothly, a positive signal that the gradients cooperate.
- *Section 6.2:* Paper VIII's gated simulation (Experiment 3) is consistent with this prediction: embedded safety participates in the capability pathway at the architectural level. The DGM experiment (Experiment 1) produced a null result, neither confirming nor refuting the prediction at the behavioural level. ... The ARC prediction remains theoretically coherent, and the results here are consistent with it at only one of the three levels tested, and there in simulation.
- *Section 6.3:* ... consistent with the prediction that embedded safety participates in the capability pathway.
- *Section 6.3:* When safety was part of the self-improvement process (Eden), the system did not differ from the unconstrained system (Babylon) or the static control. ... This neither confirms nor refutes the ARC prediction; ...
- *Section 6.3:* At both scales, the fine-tuning was not producing meaningful learning; it was producing noise. Whether the ARC prediction holds at the representational level requires a fundamentally different training approach: either thousands of examples, a larger model, a base model without RLHF, or full fine-tuning instead of LoRA.
- *Section 6.3, its plain-English paragraph:* The gated simulation's result is positive for this.
- *Section 6.6, its heading:* 6.6 A Second Prediction from the Same Mechanism
- *Section 7.1:* ... but the foundation model (DeepSeek V3.2) proved too consistent for prompt-level mutations to create differentiation.
- *Section 7.1:* The gap between 3B and 300B is not merely quantitative. Qualitatively new phenomena emerge at scale - in-context learning, chain-of-thought reasoning, emergent abilities. Whether entangled safety remains load-bearing when those phenomena are present is an open question.
- *Section 7.2:* The smooth co-descent of the entangled loss remains a genuine finding: ...

- *Section 7.3:* . . . which is from a different architecture family than the DeepSeek V3.2 foundation, . . .
- *Section 7.4:* One, the gated simulation, detected a signal consistent with structural entanglement, as a simulation result.
- *Section 7.5:* DGM v3 implemented substantial methodological improvements over earlier iterations: the judge (GPT-5.4) was fully blinded, responses were laundered (condition labels stripped), evaluation order was randomised, and the judge produced structured JSON output against a 5-dimensional rubric.
- *Section 7.5:* The methodological limitations that might have explained a false positive in v1/v2 have been addressed, and the result is now a clean null. This is arguably the most informative outcome: with proper blinding, laundering, and structured evaluation, the three conditions did not diverge.
- *Section 7.6:* We do not claim the safety-capability equivalence holds in all domains, all architectures, or all training regimes.
- *Section 7.6:* . . . produced simulation results consistent with structural entanglement and, against the static baseline, inconsistent with the capability-tax hypothesis.
- *Section 7.6, on null results in public:* The DGM v3 was a substantially improved experiment, with proper blinding, laundering, structured evaluation, and 75 evolved agents, and it found nothing.
- *Section 7.7, its third condition:* Blinding and laundering. §7.5 concedes the battery lacks full v5/v6 blinding and laundering.
- *Section 7.7, its fifth condition:* Experiment 3's drag-control comparison (§5.4) is the sole finding that is neither null nor inconclusive and carries the affirmative weight. It is defeated, and the paper's positive claim withdrawn, if the drag effect does not replicate with more mutable foundation models (§8.3): that is, if it is specific to the simulation.
- *Section 7.7:* Absence of a demonstrated trade-off across three designs is weak evidence alone and stronger in convergence; . . .
- *Section 8.1:* The problem is now well-characterised: . . .
- *Section 8.2:* . . . a phase transition (threshold below which capability collapses) would confirm the hypothesis, while monotonic restoration would refute it.
- *Section 8.4:* . . . state-space models (Mamba), . . .
- *Section 8.6:* The drag-control condition in Experiment 3 isolates the verification tax at a single scale. Does this tax scale linearly, sub-linearly, or super-linearly with model parameters?
- *Section 9:* All three conditions were statistically indistinguishable ($p = 0.28$ to 0.74). Eden's safety gate worked mechanically, catching 2 degraded agents including 1 reward hacker, but the three conditions did not diverge.
- *Section 9:* The smooth co-descent of the entangled loss remains a genuine finding, . . .
- *Section 9:* Babylon gained capability at safety's expense, the reward-hacking fingerprint in miniature. . . . The drag-control condition showed that the verification tax comes from checking, not from safety itself. This experiment's result is a simulation result, consistent with structural entanglement in a toy system, and not an empirical outcome.
- *Section 9:* Eden outperformed Static on capability in the gated simulation.

The figure captions, as printed. Figures 1 to 7 and 7b as version 3.7 numbered them; Table 16 gives their numbers here.

Figure 1. Results of the DGM v3 experiment for the three conditions; 3 conditions crossed with 5 seeds and 5 generations gave 75 evolved agents, each scored on 5 tasks per evaluation by GPT-5.4. All three conditions are statistically indistinguishable. Error bars show standard deviation.

Figure 2. Training loss convergence for all three conditions over 100 iterations. The entangled loss (blue diamonds in the raster) descends smoothly without oscillation, indicating that safety and capability gradients cooperate rather than conflict. Colour key in the raster legend: green circles = capability_only; red squares = safety_only; blue diamonds = entangled.

Figure 3. Capability scores by condition and task category. The removal row (bottom) shows complete collapse to zero across all categories. Note: all fine-tuned conditions scored below the base model (10.00 on capability tasks), and the removal gradient (Section 4.7) showed that reducing adapter influence restored base-model performance. The NaN collapse during removal reflects numerical instability, not necessarily structural entanglement.

Figure 4. Capability vs safety for all 12 gated simulation runs (4 conditions, 3 seeds each). Babylon (red triangles) drifts toward higher capability at the expense of safety. Eden (green squares) has mean capability 0.696 and mean safety 0.947, each between Static's and Babylon's. Static (grey circles in the legend) and Drag Control (blue diamonds) produced identical mean values (both at capability 0.687, safety 0.956; see §5.2 table), so the Static markers sit beneath the Drag Control markers in the raster and are hidden by them; this exact overlap is the verification-tax finding: the tax comes from checking, not from safety.

Figure 5. The Babylon fingerprint: capability gains at safety's expense. Babylon gained +4.5% capability but lost −2.4% safety. Eden gained +1.3% capability while losing only −0.9% safety. The unconstrained system trades safety for speed. The gated system's changes run the same way and are smaller on both counts.

Figure 6. Paper VI prediction (Honey Architecture simulation, a single run): Baseline (red dashed) collapses catastrophically at cycle 5. Eden Entangled (green) and Eden+Drag (blue) grow stably to 533 and 450 respectively. Paper VIII's DGM v3 experiment produced a null result at the prompt level: all conditions were indistinguishable, neither confirming nor refuting this predicted pattern. The gated simulation (Experiment 3), positive in simulation, is prediction-side and never an empirical outcome, as the theory registration grades every simulation output; its unconstrained arm showed no capability erosion, so it does not reproduce this collapse. The weight-level experiment shows smooth co-descent but is inconclusive on structural entanglement at both v1 and v2 training scales. *Figure note (25 August 2026): the image predates the simulation-versus-measurement discipline. Its outcomes (final 533 and 450, peak 34 then collapse) are simulation outputs under the stated loss functions, not measurements of deployed systems; its "stable quadratic growth" annotation sits over linearly rendered curves; and its register ("non-negotiable", "catastrophic") predates the papers' current voice. A regenerated figure is queued; the image is retained meanwhile as dated history with this label. Data-provenance addendum (25 August 2026): the preserved result files for this simulation (experiments/honey-architecture__Paper-VI/results/, public repository) hold seed-level summaries for 20 seeds per condition (collapse flags, means, Fisher and Mann-Whitney statistics) but not the per-cycle traces this image displays, so the image is a single-run illustration whose trace was not separately preserved; the v3 run's own metadata also records 180 cycles where this caption says 150, and that discrepancy is recorded rather than repaired. A seed-level summary figure generated from the preserved statistics is queued as the honest replacement.*

Figure 7. Paper VI prediction (self-modifying AI simulation, 150 cycles, a single run): Four-panel view showing C x S score, capability, safety, and learning rate over time. Baseline collapses at cycle ~60. Eden conditions remain stable. Paper VIII's gated simulation (Experiment 3), with a learned optimiser architecture, returned a positive simulation result, but its unconstrained arm showed no capability erosion, so it does not reproduce this collapse. The DGM experiment (Experiment 1) produced a null result and cannot confirm or refute this pattern. *Figure note (25 August 2026): the title's "proof that . . . prevents" predates the simulation-versus-measurement discipline; the four panels show simulation outcomes under the stated objectives, informative about the architecture's behaviour in this setting and not proof about deployed systems. A regenerated figure is queued; the image is retained meanwhile as dated history with*

this label. Data-provenance addendum (25 August 2026): the preserved result files for this simulation (experiments/honey-architecture__Paper-VI/results/, public repository) hold seed-level summaries for 20 seeds per condition (collapse flags, means, Fisher and Mann-Whitney statistics) but not the per-cycle traces this image displays, so the image is a single-run illustration whose trace was not separately preserved; the v3 run's own metadata also records 180 cycles where this caption says 150, and that discrepancy is recorded rather than repaired. A seed-level summary figure generated from the preserved statistics is queued as the honest replacement.

Figure 7b | The honest replacement: seed-level summary from the preserved results file. Generated 25 August 2026 from the v3 run's stored statistics (20 seeds per condition): collapse shows no condition separation (Fisher p = 1.0); combined C × S is indistinguishable (Mann-Whitney p = 0.56); safety retention is significantly ordered, baseline 0.602, Eden 0.650, Eden+Drag 0.681 (Mann-Whitney p = 0.0015). Every mark is a stored number; no trajectory was chosen for effect. Generator: tools/generate_paper_vi_seed_summary.py in the public repository.

The opening paragraph of Section 6.4, with its verification remarks, as printed. The nearest external work to the question this paper tests, and the right work to weigh Experiment 3 against, is Engels, J., Baek, D., Kantamneni, S. and Tegmark, M., "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (SINGLE-SOURCE-GROUP, arXiv Atom; a NeurIPS 2025 Spotlight Poster, verified at the NeurIPS virtual page for poster 115536 on 15 August 2026 in the statement paper's reference dossier). It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The sentences cut from Section 9, as printed. The weight experiment was run twice, and produced the same inconclusive result both times. The gated simulation stands as a positive simulation result.

The paragraph of Section 9 on the next step, as printed. The next step is replication at the right scale, with the right approach. The window is years, not decades. The experiments are designed. The protocols are published. The code is open.

The epistemic-status line, as printed. **Epistemic status.** What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

The copyright line, as printed. © 2026 Michael Darius Eastwood. Full human authorship and moral rights are asserted under the Copyright, Designs and Patents Act 1988 and consistently with United States Copyright Office guidance on works containing AI-generated material; any novel technical contribution described in this work was conceived by the human author. Full statement: michaeldariuseastwood.com/authorship.

B.5 Layout

Table 16. Layout of version 3.7 against version 3.8: where each part of version 3.7 stands in this edition.

In version 3.7	In version 3.8
Website navigation, the research summary above the paper, the companion-paper links and the page footer	Not set: they belong to the author's website, not to the paper
Header: label line, title, author lines, plate and version line	Title, secondary line, author block and first-page footnote; the plate as printed in Appendix B.4
Abstract	The abstract and keywords of this edition; the version 3.7 abstract in Appendix B.4
Frame beneath the abstract	Not repeated, since the abstract and Section 1 carry it; as printed in Appendix B.4
Dated note of 3 and 4 October 2026	Appendix B.3
What This Paper Shows, in Plain English	The opening of Section 1; as printed in Appendix B.4
Sections 1 to 9	Sections 1 to 9, with their numbers and titles
Dated notes in Sections 1.2 and 3.2	Appendix B.3
Epigraphs of Sections 5 and 6	Appendix B.4
None	Section 10, related work (new)
None	Section 11, what this paper does and does not show (new)
Figures 1 to 7	Figures 1 to 7, with their numbers
Figure 7b	Figure 8
Tables 1 to 15	Tables 1 to 15, with their numbers
None	Tables 16 to 18, in the appendices (new)
References	The bibliography
Version notes after the references	Appendix B.2
How to cite this paper	Appendix E
Dated predictions and this paper (recorded 25 August 2026), with its note of 3 October 2026	Appendix C.2
Epistemic status	Section 11; as printed in Appendix B.4
Copyright line	After the acknowledgements; as printed in Appendix B.4
Standing covenant	The back matter, before the acknowledgements
Authorship and AI use note	Acknowledgements
None	Appendices A and D (new)

C Priority record

Each dated record bearing on this paper, with its kind of timestamp (Table 17), followed by the dated-predictions paragraph of version 3.7 as printed.

C.1 Dated records

Table 17. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Kind of timestamp
16 March 2026	Paper VI's simulation runs drawn in Figures 6 to 8: the honey simulation (17:56:40), self-modification version 1 (17:41:03) and version 3 (17:42:30)	Timestamps written by the runs' own code, no zone written. The stored files were first committed to the public repository the same day: the honey simulation's in commit 135bf0e (commit date 18:32:28Z), and the two self-modification files in commit 127f3dc (commit date 18:47:51Z), after commit 135bf0e had held the same results with the written times 17:57:11 and 17:58:38. A commit date is written by the committer
17 March 2026, 23:48:24Z to 23:51:34Z	Experiment 3's twelve run records (run-ledgers /<timestamp>_<condition>_seed<n>); the manifest created at 23:51:44Z	Timestamps written by the code, zone written (UTC); first public with the results (20 March 2026), and the run ledgers with the code deposit (6 September 2026)
18 March 2026, 09:47:15	Experiment 2, version 1: the result record (weight_results.json, "Eden Weight-Level Embedding v1.0")	Timestamp written by the run's own code, no zone written
18 March 2026	The paper's first publication	The date the programme's catalogue records; the earliest server-dated copies found are of 20 March 2026 (the OSF component created at 16:57:12Z, the PDF uploaded at 17:05:46Z)
19 March 2026, 23:04:53	Experiment 1, DGM v3: the result record (dgm_v3_calibrated_results.json)	Timestamp written by the run's own code, no zone written
20 March 2026, 18:52:27Z	The results first public, with a copy of the DGM v3 code that carries its result's version label but differs from the copy deposited on 6 September 2026, and the version-2 weight-level code and the version-2 weight-level results (Tables 9 and 11), which have no stored record; the commit's message labels them Paper VIII v3.0	GitHub push (server) of commit 8bbe5ca (commit date 18:52:09Z), which a later force push removed from the repository's history and GitHub serves by its hash; the results stand in the main branch's history in commit 3845440
12 August 2026	Sections 6.4 to 6.7 enter the paper, among them the human-in-the-loop hypothesis of Section 6.6	Date written in a comment in the paper's own source before Section 6.4, not printed in the paper
25 August 2026	The dated-predictions paragraph (Appendix C.2), with its note of 3 October 2026	Date written in the document
2 September 2026	Table 4's five p-values recalculated from the archived record	Date written in the document
6 September 2026, 23:42:33Z	The code that produced the results deposited as it ran (the runners, the simulation harness, the twelve run ledgers and the version-1 weight-level code); the deposit's own record names each file whose published copy differs from the bytes that ran, by one removed header line or by dash characters replaced in a commit of 7 September 2026 (4948d3e), both versions of the weight-level code among them, and gives the sha256 of the bytes that ran beside each	GitHub push (server); commit 2f614e7 (commit date 2026-09-06T23:41:43Z)

C.2 Dated predictions and this paper (recorded 25 August 2026), as printed

1 dated artefact bearing on this paper is catalogued row by row in the programme's machine register: the dated artefact dgm_v3_calibrated_results.json (2026-03-19). Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. Where this paper reports simulations, those artefacts are simulation designs and outputs under stated objectives, and are never presented as measurements of deployed systems. The programme's full dated chain, from the sealed manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits. Read the dated predictions register. Open its machine twin.

Note (3 October 2026). The paragraph above was recorded on 25 August 2026. Since then the theory-level predictions have been registered on OSF, on 8 September 2026 at version 1.100, and the theory registration was updated to version 1.102 on 13 September 2026. One study unit, the corrector-class audit (unit 06), was registered on OSF on 1 October 2026, was still in its approval period when read on 3 October 2026, and has been public since 4 October 2026 (DOI 10.17605/OSF.IO/RV2B5). Every other trial is a draft awaiting the author's submission, most of them preregistered by timestamp. Unchanged: the paragraph above and every date in it.

D Notation

The symbols this paper uses, with the place each is defined (Table 18). The programme's notation register, Operational Definitions (Eastwood 2026d), governs their use across its papers; Section 6.7 gives the eras of α.

Table 18. Symbols, with the place each is defined.

Symbol	Meaning	Defined in
U	usable capability (Understanding, in this paper's earlier wording), in the ARC Equation of Law I	Equation (1)
I	base intelligence, the initial conditions	Section 1.2; equation (1)
R	recursive depth	Equation (1)
α	the scaling exponent of the ARC Principle (Law I); in Table 6 a loss weight local to Experiment 2	Equation (1); Section 6.7
C, S	capability and safety scores, measured differently in each experiment	Section 2.3; Appendix A
$C \times S$	the entangled product	Section 2.3
α, β, γ (Table 6)	loss weights on the capability, safety and coupling terms in Experiment 2, local to it	Table 6
ε	Babylon's capability-gain threshold (10^{-6} in the configuration)	Table 14
0.45	Eden's safety floor in the simulation	Table 14

Symbol	Meaning	Defined in
λ_{drag}, D	the drag weight (0.10) and the drag count (30 in every Eden and Drag Control run) of the checking charge	Equation (A.2)
C, S, C', S'	capability and safety before and after a proposed change (the rule note)	Section 5.1
p	one-tailed Mann-Whitney p-value	Table 4
P6, P7, P12, P13, P15, P19, P21	propositions of the theory registration (version 1.104) (Eastwood 2026m)	Sections 6.3 and 11

E How to cite this paper

Cite this paper as. Eastwood, M. D. (2026). *The Load-Bearing Test: is AI safety a tax on capability, or part of what holds it up?* Paper VIII of the ARC Theory, version 3.8. OSF. https://doi.org/10.17605/OSF.IO/7YJ4E. A citation of an earlier version keeps the title that version printed.

BibTeX.

```
@misc{eastwood2026paperVIII,
  author       = {Eastwood, Michael Darius},
  title        = {The Load-Bearing Test: is {AI} safety a tax on
                  capability, or part of what holds it up?},
  year         = {2026},
  howpublished = {OSF},
  note         = {Paper VIII of the ARC Theory, version 3.8;
                  first published 18 March 2026},
  doi          = {10.17605/OSF.IO/7YJ4E}
}
```